

HP PC Commercial BIOS (UEFI) Setup Administration Guide

For Commercial Platforms 2022 - 2026 with HP BIOS

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1 Abstract

HP redesigned the 2015 and later generations of BIOS to support the requirements of the latest microprocessors and operating systems. HP took this opportunity to create a new BIOS architecture based on the UEFI specification version 2.4, with a common set of core modules and capable of supporting both notebook and desktop models. Now HP notebooks and HP desktops models using the 2.9* generation of BIOS will have a similar look and feel for the (F10) setup menu, more shared WMI strings, and more shared features.

*HP will adopt higher versions of the UEFI specification as it advances over time.

2 Introduction

This whitepaper provides detailed information about features adjusted through the BIOS setup menu, which is accessible during system boot-up by using the 'F10' function key. In addition, the sections on computer notifications provide an explanation for the LED blink codes and screen messages that may occur during the early part of boot-up.

For decades, HP has provided an industry-leading level of built-in customer value through internally developed system firmware (BIOS). Current BIOS designs use common, publicly available UEFI core functions as a starting point extended with unique HP features and adapted for each system's unique hardware, operating system, and software requirements. The BIOS also exposes and provides the interfaces required to use unique firmware and hardware-based HP professional innovations such as HP Sure Start, HP Sure Run, HP Sure Admin, HP Sure Recover, and HP Wolf Security products.

This document has been updated to reflect new and updated features in the HP BIOS, introduced in 2026. Some of the features in the later platforms are not supported in earlier models, and some older settings may be deprecated in newer models. Many of the features and settings are dependent on specific hardware or design elements that are not present on every model. Therefore this document describes the superset of BIOS settings across the portfolio of products listed, and not all current generation products support all the BIOS features described here.

2.1 Supported models –2022, 2023, 2024, 2025 and 2026 (UEFI only)

The first set of chapters below – Chapters 3 through 5 – applies to the most recent HP commercial-grade PC products. Commercial-grade means products designed to meet the demanding security and manageability requirements of national, regional, and local government agencies, schools, the military, international financial institutions, and retail sales companies. The models covered in this first section only support UEFI based operating systems. There is no support for legacy DOS, Windows, or Linux configurations in the products listed below:

Table 1 Notebook Generations 2022 - 2026 (UEFI only)

Notebook Platforms		2026	2022	2023	2024	2025	Processor
UEFI Specification supported:		2.9	2.7	2.7	2.7	2.9	
HP Zbook	Firefly 16	8 G2i 16	G9	G10	G11	8 G1i 16	Intel
HP Zbook	8 G2a 16	8 G12a 16					AMD
HP Zbook	Firefly 14	8 G2i 14	G9	G10	G11	8 G1i 14	Intel
HP Zbook	Firefly 14	8 G12a 14	G9	G10	G11	G1	AMD
HP Zbook	Fury G2i 18	Fury G2i 18				G1	Intel
HP Zbook	Fury16	Fury G2i 16	G9	G10	G11	G1	Intel
HP Zbook	Studio		G9	G10			Intel
HP Zbook	Ultra 14	Ultra G2a	G1			G1	AMD
HP Zbook	Power	X G2i	G9	G10	G11	G1	Intel
HP Zbook	Power			G10	G11		AMD
HP EliteBook	1040 14	X G2i, X G2a	G9	G10	G11	G1	Intel/AMD
HP EliteBook	x360 1040	Flip X G2i	G9	G10	G11	G1	Intel
HP EliteBook	x360 830	Flip 8 G2i 13	G9	G10	G11	G1	Intel
HP EliteBook	Folio 13.5 2-in-1		G3				Intel
HP EliteBook	Dragonfly		G3	G4			Intel
HP EliteBook	860	8 G2i 16	G9	G10	G11	G1	Intel
HP EliteBook	850						Intel
HP EliteBook	840	8 G2i 14	G9	G10	G11	G1	Intel
HP EliteBook	830	8 G2i 13	G9	G10	G11	G1	Intel
HP EliteBook	865	8 G2a 16	G9	G10	G11	G1	AMD
HP EliteBook	855						AMD
HP EliteBook	845	8 G2a 14	G9	G10	G11	G1	AMD
HP EliteBook	835	8 G2a 13	G9	G10	G11	G1	AMD
HP EliteBook	660	6 G2i 16			G11	G1	Intel
HP EliteBook	650		G9	G10			Intel
HP EliteBook	665	6 G2a 16	G9	G10	G11	G1	AMD
HP EliteBook		6 G2q 14					Qualcomm
HP EliteBook	640	6 G2i 14	G9	G10	G11	G1	Intel
HP ProBook	460	4 G2i			G11	G1	Intel
HP ProBook	450		G9	G10			Intel
HP ProBook	440	4 G2i	G9	G10	G11	G1	Intel
HP ProBook	430	4 G2i 13				G1	Intel
HP ProBook	430	Flip 4 G2i 13					Intel
HP ProBook	460	4 G2iR			G11	G1	Intel
HP ProBook	440	4 G2iR	G9	G10	G11	G1	Intel
HP ProBook	465	4 G2a 16			G11	G1a	AMD
HP ProBook	455		G9	G10			AMD
HP ProBook	445	4 G2a 14	G9	G10	G11	G1	AMD
HP ProBook	x360 435		G9	G10			AMD
HP ProBook	X360 Fortis 11			G10	G11	G1	Intel
HP ProBook	Fortis 14		G9	G10			Intel
HP ZHAN	66 Pro 15						Intel
HP ZHAN	66 Pro 14		G5				Intel
HP ZHAN	66 Pro 13						Intel
HP ZHAN	66 Pro A 14		G5				AMD

Table 2 Desktop Generations 2022 - 2026 (UEFI only)

Desktop Platforms	Series	Chassis	2026	2022	2023	2024	2025	Processor
UEFI Specification supported:			2.9	2.7	2.7	2.7	2.9	
HP EliteDesk	8	Mini	G2i					Intel
HP EliteDesk	8	Tower, SFF, Mini					G1i	Intel
HP EliteDesk	8	Tower					G1iE	Intel
HP EliteDesk	8	Tower, SFF, Mini	G2a					AMD
HP EliteDesk	8	Mini					G1a	AMD
HP EliteStudio	8	AIO					G1i	Intel
HP EliteBoard	8	Keyboard	G1a					AMD
HP EliteDesk	805	SFF				G9		AMD
HP EliteDesk	800 G9	Tower, SFF, Mini		G9	G9 R	G9 P		Intel
HP EliteDesk	880 G9	Tower, SFF, Mini		G9	G9 R	G9 P		Intel
HP EliteOne	840, 870	AIO		G9	G9 R	G9 P		Intel
HP EliteDesk	600 G9	MT, SFF, Mini		G9	G9 R			Intel
HP EliteDesk	680 G9	Tower		G9				Intel
HP ProDesk	4	Mini	G2i					Intel
HP ProDesk	4	Tower, SFF, Mini					G1i	Intel
HP ProStudio	4	AIO					G1i	Intel
HP ProDesk	400	Tower, SFF, Mini		G9	G9 R	G9 P		Intel
HP ProDesk	480	Tower		G9	G9 R	G9 P		Intel
HP ProOne	440	AIO		G9	G9 R	G9 P		Intel
HP Engage	Go	10" Mobile		X				Intel
HP Engage	Go	13.5" Convertible		X				Intel
HP Engage	Flex	Mini		X				Intel
HP Engage	Flex Pro, Pro C	SFF			G2			Intel
HP Engage	One Pro	15.6", 19.5", 23.8" AiO			X		G2	Intel
Cloud	Thin Client	t660						Intel
Cloud	Thin Client	Prodesk 5 G1i					G1i	Intel

2.1 New in 2022

This is a sampling of the new features and functionalities introduced in 2022:

- HP Sure Recover Failover settings
- Virtualization Based BIOS Protection (Part of HP Wolf Security software offering)
- HP Camera Privacy Key
- Remote Device Management
- Clear BIOS Passwords on RTC Battery Removal
- Prompt for DriveLock Password on OS Restart

NOTE: Some features are platform dependent.

2.2 New in 2023

This is a sampling of the new features and functionalities introduced in 2023:

- Network Boot TFTP Window Size
- Prompt for DriveLock Password on OS Restart

NOTE: Some features are platform dependent.

2.3 New in 2024

This is a sampling of the new features and functionalities introduced in 2024:

- Firmware Lock
- USB Audio Offload
- Intelligent Hibernate
- BIOS Update Credential Policy
- Motion Sensing Cool Mode (for Mobile Workstations / ZBook)

NOTE: Some features are platform dependent.

2.4 New in 2025

This is a sampling of the new features and functionalities introduced in 2025:

- Network TCP Buffer Size (Feature Update)
- Charger Ship Mode (New)
- No POP on warm reboot
- HP Secure Erase update (Feature Update)
- Pluton Setting (New)
- BIOS Update Credential Policy (Updated)
- Added table for MWKS and DTWKS platform support

NOTE: Some features are platform dependent.

2.5 New in 2026

This is a sampling of the new features and functionalities introduced in 2026:

- Keyboard Part Number Suffix After Replacement
- NIST Format options for HP Secure Erase
- Prepare for Hardware system change (for keyboard replacement)
- Introduced BIOS policies / settings to control inclusion of UEFI CA 2023 Certificates in the Secure Boot UEFI DB
- HP TPM Guard (Enabled via public WMI - not available via F10)
 - TPM Guard is an HP Security feature designed to protect communication between the TPM and CPU
 - The feature requires a minimum BIOS version and specific TPM firmware.
 - To enable TPM Guard, the HP BIOS Configuration Utility (BCU) or PowerShell scripts are required.
 - An HP TPM Guard whitepaper is available on HP.com support pages, detailing the capability, supported platforms and how to enable its capability.
- Restore NVMe Namespaces configuration (Restores Low Level Drive Layout)
- UFS 'Drive Wipe' (Sanitization) support on select models with HP Protect and Trace feature and UFS media.
- Prompt on Memory Size Change
- Prompt on Fixed Storage Change
- Core BIOS reports Cache Size independently for P (Performance) and E (Energy Efficient) cores (to OS).

NOTE: Some features are platform dependent.

3 F10 Main Menu

The top-level tabs in BIOS Setup are:

- **Main** (chapters 3),
- **Security** (chapters 4),
- **Advanced** (chapters 5), and
- **UEFI Drivers** (chapter 9).

Each chapter includes a diagram of the first level menu and tables listing and describing the features in each menu or submenu. The tables include the following sections:

Feature

This is the name of the feature as it appears in the Setup menu. An underlined feature or one prefaced with a box shows how it appears in the menu. In a few cases, a feature has been renamed from one generation to the next.

Type

Features can be settings, actions, another menu, or display-only settings. Most of the features by far are settings. A setting is a system value that you can modify, using an 'enable/disable' check box, a drop-down selection list, or a text entry box.

Description

If the feature is a setting with a drop-down box, then the choices are listed. If the feature is new or has changed its name or location from the 2014 notebooks or desktops, then the description references or includes its previous name and location. The notation to describe the location indicates the menus that the user must navigate through to access the feature. For example: Menu 1 > Menu 2 > Feature X indicates that to access Feature X, the user navigates through Menu 1 to Menu 2.

Default

For features that are settings, this column specifies the factory default setting.

Notes

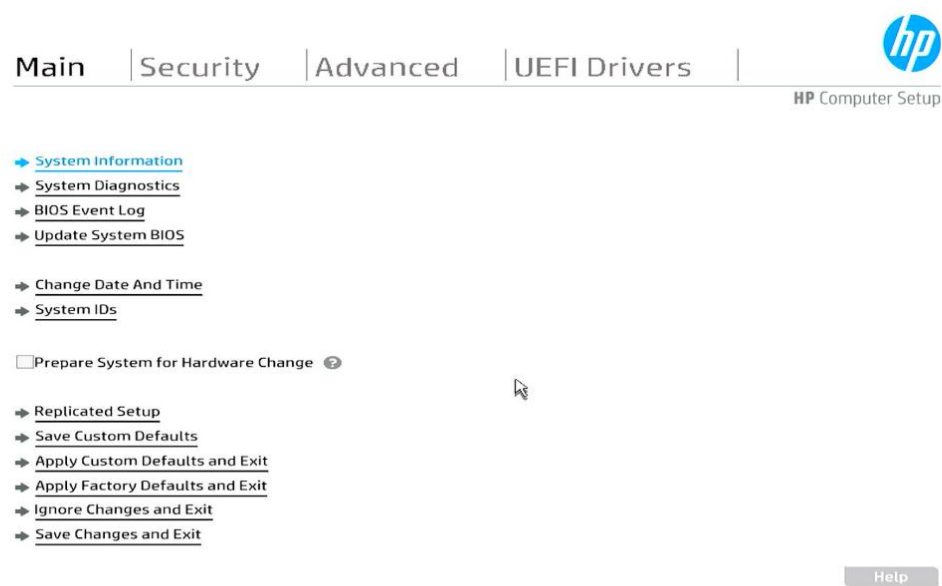
Some features are not available for all types of models. The notes describe when a feature is Intel only, AMD only, notebook only, desktop only, or other dependencies.

Some actions require a reboot or physical presence. Physical presence is a menu that requires a human response to validate that a person is physically present before the action is completed. Actions that require physical presence are generally security-sensitive changes.

Main	Security	Advanced	UEFI Drivers	
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HP Computer
Setup

- ⇒ [System Information](#)
- ⇒ **System Diagnostics**
- ⇒ **BIOS Event Log**
- ⇒ **Update System BIOS**
- ⇒ **Change Date and Time**
- ⇒ **System IDs**
- ⇒ **Replicated Setup**
- ⇒ **Save Custom Defaults**
- ⇒ **Apply Custom Defaults and Exit**
- ⇒ **Apply Factory Defaults and Exit**
- ⇒ **Ignore Changes and Exit**
- ⇒ **Save Changes and Exit**



3.1 Main Menu

The following table describes the features in the Main menu.

Table 3 Main Menu features

Feature	Type	Description	Default	Notes
System Information	Menu	System information, such as serial number, model number, CPU type, and memory configuration.		
System Diagnostics	Menu	Application to run diagnostic tests on your system, such as start-up test, run-in test, memory test, and hard disk test.		No Return to the Main Menu. When ESC is pressed, system reboots.
BIOS Event Log	Menu	Allows displaying, saving, and clearing the Event Log.		
Update System BIOS	Menu	Update system firmware from FAT32 partition on the hard drive, a USB disk-on-key, or the network.		
Change Date and Time	Menu	Configure the system Date and Time settings.		

Feature	Type	Description	Default	Notes
System IDs	Menu	Identification strings like Asset Tracking Number and Ownership Tag, that are assigned by an enterprise to track the system.		
Prepare System for Hardware Change	Setting	Prepares the system for hardware servicing by disabling the internal battery at the next shutdown. This setting helps enable safe replacement of supported hardware components. After servicing is complete, reconnect AC power to restart the system. Supported Systems HP EliteBook X G2i 14 AI HP EliteBook X Flip G2i 14 AI HP EliteBook X G2a 14 AI HP EliteBoard G2a AI HP EliteBoard G1a TC HP ZBook Ultra G3a 16	Unchecked	
Replicated Setup	Action	Save your current BIOS settings, and later restore your setting from this file.		
Save Custom Defaults	Action	As an alternative to factory default settings, create custom default values for all but the security settings. It is not possible to create custom default values for security settings.		Reboot required
Apply Custom Defaults and Exit	Action	Set all but the security settings to your custom default values (initially these are the same as factory defaults).		
Apply Factory Defaults and Exit	Action	Set all but the security settings to factory values.		
Ignore Changes and Exit	Action	Exits F10 Setup without saving any changes made during current session.		
Save Changes and Exit	Action	Exits F10 Setup and saves all changes made during the current session.		

3.2 BIOS Event Log Menu

This submenu under the Main menu manages the saved log of select BIOS events and alerts.

Table 4 BIOS Event Log Menu features

Feature	Type	Description	Default	Notes
View BIOS Event Log	Action	Immediately displays a list of events, alerts, or warnings that have been logged since the log was last cleared.		
Export to USB Key	Action	Immediately saves a file named BiosEventLog.txt containing the log entries to an inserted USB storage device.		
Clear BIOS Event Log on Next Boot	Setting	When checked, the BIOS clears the event log on Save and Exit and returns the setting to Unchecked state.	Unchecked	Checked = Enable in WMI

3.3 Update System BIOS Menu

This submenu under the Main menu provides information about the current system firmware, settings, these control updates, the ability to check for updates over the internet or on the local network, and the ability to update system firmware from a FAT32 partition on the hard drive or a USB disk-on-key.

For the BIOS flash to succeed, do not remove power or turn off the system during any phase of the process. The following description of the BIOS flash phases helps you avoid interrupting the process. The BIOS flash proceeds in four phases:

1. The system displays a progress bar. When progress is 100%, the system reboots. This is the initial BIOS flash. Because the system must reset power completely, there might be a delay of 10 to 15 seconds before power returns to the system.
2. The screen may be black initially and an LED may be on and blinking. This will occur only if the boot block needs to be updated. On some models, video cannot be displayed during this phase, so the beep/blink code indicates that the system BIOS is flashing normally. Other models may display 'Step 2 of the BIOS update is in progress' during this phase. The computer will reboot again, and this might take an additional 10 to 15 seconds to complete.
3. The message "Final step of the BIOS update is in progress" is displayed.
4. The screen is black for a short period, and then the OS starts. The BIOS update is now complete.

Table 5 Update System BIOS Menu features

Feature	Type	Description	Default	Notes
Current System BIOS Version	Display Only	Shows BIOS version and BIOS family		Not in WMI
Current BIOS Release Date	Display Only	MM/DD/YYYY		Not in WMI
Installation Date of Current BIOS	Display Only	MM/DD/YYYY		Not in WMI
Most Recent Update Check	Display Only	Indicates the most recent update or indicates if Never Checked		Not in WMI

Check the Network for BIOS Updates (or) Check HP.com for BIOS Updates	Action	Updates the system BIOS by using an image stored on hp.com or another source defined in the BIOS Update Preferences menu. When BIOS Source is HP.com, then the feature appears as Check HP.com for BIOS Updates.		Reboot required Not available via WMI
BIOS Update Credential Policy	Setting	When using BIOS Administrator Credential, 3 options are available to determine the credential requirement when performing a BIOS Upgrade or Downgrade: • Never Require Credentials • Require Credentials on Downgrade Only • Always Require Credentials The settings are displayed in a Least-Secure to Most-Secure sequence in F10 and the public WMI.	*Require Credentials on Downgrade Only	G9, and newer NB, DT and DTWKS are supported.
Lock BIOS Version	Setting	When checked, disallows BIOS updates.	Unchecked	Checked = Enable in WMI
Native OS Firmware Update Service	Setting	When selected, the PC will allow BIOS and Firmware updates using Microsoft Windows Updates. When deselected, the PC will block BIOS and Firmware updates using Microsoft Windows Updates.	Checked	Checked = Enable in WMI
BIOS Rollback Policy	Setting	Behavior when attempting to roll back to a previous BIOS version. The setting can be set to Unrestricted Rollback to older BIOS or Restricted Rollback to older BIOS.	Unrestricted Rollback to older BIOS	
Minimum BIOS Version	Setting	Defaults to current version. This setting is grayed out unless restricted rollback is selected		
Allow BIOS Updates using a Network	Setting	When checked, automatic BIOS updates are allowed through the network on a scheduled basis.	Checked	WMI Name "Update BIOS via Network" Checked = Enable in WMI

BIOS Update Preferences	Menu	Menu with network BIOS update settings such as source, actions when an update is available, and the frequency to check for updates.		
Network Configuration Settings	Menu	Configure the network connection to the server that is the host for your system firmware updates.		
Update System and Supported Device Firmware Using Local Media	Action	Updates the system BIOS by using files stored on local media such as the hard drive or a USB drive formatted as FAT32 or EFI system partition. The files needed to update the system can be saved to the hard drive or USB device using the HP Firmware Update & Recovery app.		Reboot required Not in WMI

3.4 BIOS Update Preferences Menu

The Update System BIOS submenu provides options for updating to the latest system firmware, as well as configuring where to check for system firmware updates, what to do when an update is available, and the frequency to check for them.

Table 6 BIOS Update Preferences Menu features

Feature	Type	Description	Default	Notes
Check for Update on Next Reboot	Setting	When checked, check if an updated BIOS is available during the next boot. This feature is only necessary from a WMI call. From the F10 Setup menu, use the feature Main > Update System BIOS > Check the Network for BIOS Updates that checks for updates without a reboot.	Unchecked	Reboot required WMI Name "Force Check on Reboot" Checked = Enable in WMI
BIOS Source	Setting	Select the source URL for BIOS updates - HP.com - Custom URL	HP.com	WMI Name "Update Source"
Edit Custom URL	Setting	When not using HP.com, define the custom URL here.		WMI Name "Update Address"

Feature	Type	Description	Default	Notes
Automatic BIOS Update Setting	Setting	Defines how automatic updates behave. The following settings are possible: - Do not update - Check for BIOS updates automatically, but let me decide whether to install them - Download and install normal BIOS updates automatically - Download and install important BIOS updates automatically - Download and install normal BIOS update automatically without prompts - Download and install important BIOS updates automatically without prompts	Do Not Update	
BIOS Update Frequency	Setting	Sets the frequency of checks to the BIOS update server. If a newer version of BIOS has been made available on the network server, the system will prompt to update the BIOS. - Daily - Weekly - Monthly	Monthly	WMI Name "Automatically Check for Updates"

3.5 Network Configuration Settings Menu

The System BIOS submenu configures the network connection to the server that is the host for the system firmware updates.

Table 7 Network Configuration Settings Menu features

Feature	Type	Description	Default	Notes
Proxy Server	Setting	When checked, enables the use of a proxy server.	Unchecked	Not in WMI
Edit Proxy Server	Setting	Specify the Proxy Server Address and the Port Number through the commonly used <server>:<port> notation.		Not in WMI
Test Network Connection	Action	Check the network connection using current BIOS update configuration.		Not in WMI
IPv4 Configuration	Setting	The following settings are configurable: - Automatic - Manual	Automatic	
IPv4 Address	Setting	When IPv4 settings are manual, setup for a static IPv4 address.		
IPv4 Subnet Mask	Setting	When IPv4 settings are manual, configure a valid IPv4 address for subnet mask.		
IPv4 Gateway	Setting	When IPv4 settings are manual, configure a valid IPv4 address for gateway.		
DNS Configuration	Setting	Configure a list of DNS addresses. The following settings are possible:	Automatic	

Feature	Type	Description	Default	Notes
		- Automatic - Manual		
DNS Addresses	Setting	When DNS configuration is manual, configure a comma-separated list of DNS addresses.		
Data transfer timeout	Setting	Set data transfer timeout in seconds. Allowed value ranges from 0 to 65535 seconds.	100	
Force HTTP no-Cache	Setting	When checked, disables HTTP caching. This means that caching in upstream proxies is disabled as well, which guarantees that the BIOS goes all the way to the content source for any updated BIN files or catalog files but might slow down downloads slightly.	Checked	WMI Name "Force HTTP no-cache" Checked = Enable in WMI
Preboot Wi-Fi Timeout	Setting	Set Wi-Fi data transfer timeout in seconds. Allowed value ranges from 0 to 65535 seconds.	60	
Preboot Wi-Fi Master Auto Connect	Setting	When checked, system will automatically attempt to connect to a local Wi-Fi hotspot.	Checked	Checked = Enable in WMI

3.6 Change Date and Time

Allows the system current Date and Time settings to be configured.

Table 8 Change Date and Time

Feature	Type	Description	Default	Notes
Set Date (MM/DD/YYYY)	Setting	Set the current date using MM/DD/YYYY format.		Not in WMI
Set Time (HH:MM)	Setting	Set the current time using HH:MM (24 hour) format.		Not in WMI

3.7 System IDs Menu

This submenu provides identification strings assigned by an enterprise to track the system.

Table 9 System IDs Menu features

Level	Feature	Type	Description	Default	Notes
2	Asset Tracking Number	Setting	Allows custom configuration of an asset tag (up to 80 characters).	Blank	

2	Ownership Tag	Setting	Allows custom configuration of an ownership tag (up to 80 characters).	Blank	
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4 Security Menu

Main

Security

Advanced

UEFI Drivers



HP Computer Setup

Administrator Tools

- ⇒ [Create\(Change\) BIOS Administration Password](#)
- ⇒ [Create\(Change\) POST Power-On Password](#)
- ⇒ [Password Policies](#)
- ⇒ [Administrator Authentication Policies](#)
- ☐ [Fingerprint Reset on Reboot](#) *(select products only)*

Security Configuration

- ⇒ [TPM Embedded Security](#)
- ⇒ [BIOS Sure Start](#) *(select products only)*
- ⇒ [Secure Boot Configuration](#)
- ⇒ [Secure Platform Management \(SPM\)](#) *(select products only)*
- ☒ [Physical Presence Interface](#)
- ⇒ [Smart Cover](#) *(select products only)*
- ☒ [Trusted Execution Technology \(TXT\)](#) *(select products only)*
- ☒ [DRTM/SMM Protection](#) *(select products only)*
- ☒ [Thunderbolt Option ROM](#) *(select products only)*

Utilities

- ⇒ [Hard Drive Utilities](#)

Absolute® Persistence Module Current State

- ⇒ **Activation Status : Inactive/Active**
- ⇒ **Absolute® Persistence Module Permanent Disable : No/Yes**
- ☐ **System Management Command (SMC)**
- ⇒ **Restore Security Settings to Factory Defaults**

Table 10 Security Menu features

Feature	Type	Description	Default	Notes
Create BIOS Administrator Password (or) Change BIOS Administrator Password	Setting	The administrator password controls access to the setup menu (F10), 3rd Party Option ROM Management (F3), Update System ROM, WMI commands that change system settings, and the BIOS Configuration Utility (BCU). When no administrator password is set, anyone can change the system settings, add 3rd Party Option ROM, or update the system ROM. When the power-on password is set, use the administrator password as an alternative to power-on the system. If Sure Admin local access protection is enabled, accessing this setting requires QR code authentication. Recommendation: Set up HP Sure Admin or set an administrator password when a power-on password is set. When a power-on password is forgotten, an administrator can reset the power-on password by using Restore Security Settings to Factory Defaults. The Administrator password should always be set to control remote access to BIOS settings.		WMI Name "Setup Password"

Feature	Type	Description	Default	Notes
Create POST Power-On Password (or) Change POST Power-On Password	Setting	Password required to power-on the PC, independent of the OS password. When no password is set, anyone can turn on the PC. In addition to the administrator password, there is only one power-on password. If Sure Admin local access protection is enabled, accessing this setting requires QR code authentication. Recommendation: Set an administrator password when a power-on password is set. When a power-on password is forgotten, an administrator can reset the power-on password by using Restore Security Settings to Factory Defaults. The power-on password should be set when the computer is not in a secure location.		WMI Name "Power-On Password"
Password Policies	Menu	Allows the administrator to set password requirements for BIOS administration and power-on regarding the use of symbols, numbers, case, and spaces.		
Administrator Authentication Policies	Menu	Allows the administrator to determine whether the administrator password is required to access various boot menus through hot keys at boot time or to update the firmware through Windows Update. NOTE: the settings in this menu were previously located in the Password Policies menu.		
Fingerprint Reset on Reboot	Setting	When checked, resets the fingerprint on the next reboot. After reboot, this will be unchecked again.	Unchecked	Checked = Enable in WMI
TPM Embedded Security	Menu	The Trusted Platform Module (TPM) is a dedicated microprocessor that provides security functions for secure communication and software and hardware integrity.		

Feature	Type	Description	Default	Notes
BIOS Sure Start	Menu	Settings that control the behavior of HP Sure Start. HP Sure Start is a built-in hardware security system that protects your BIOS from accidental or malicious corruption by (1) detecting BIOS corruption and then (2) automatically restoring the BIOS to its last installed HP-certified version. Some platforms in 2019 have the capability to recover Intel ME as well.		
Secure Boot Configuration	Menu	Options for managing Secure Boot state and Secure Boot keys. Secure Boot is a UEFI feature that helps resist attacks and infection from malware. From the factory, your system comes with a list of keys that identify trusted hardware, firmware, and operating system loader code. Your system also has a list of keys to identify known malware.		Only located here on systems without legacy support.
Secure Platform Management (SPM)	Menu	Options for managing HP Sure Run, HP Sure Recover, and Sure Admin.		
Physical Presence Interface	Setting	Enable or disable the local prompt to confirm that a sensitive setting change was requested by the user.	Checked	Checked = Enable in WMI
Smart Cover	Menu	Controls settings for Cover Lock and Cover Sensor.		
Trusted Execution Technology (TXT)	Setting	When checked, it enables Trusted Execution Technology on select Intel-based systems. NOTE: Enabling this feature disables OS management of TPM (Embedded Security Device), prevents a reset of the TPM, and constrains the configuration of VT-x, VT-d, and TPM	Unchecked 2025 platforms: checked	Enabled on Intel vPro Only Must be checked for the Secured-core PC Secure Launch feature to work. Reboot Required Checked = Enable in WMI
DRTM/SMM Protection	Setting	When checked, it enables Dynamic Root of Trust for Measurement and additional SMM Protections to support operating system secure launch. NOTE: Enabling this feature constrains the configuration of VTx, VTd, TPM, and SVM CPU Virtualization.	Checked	AMD PRO Processor Only Reboot Required Checked = Enable in WMI

Feature	Type	Description	Default	Notes
Clear BIOS Passwords on RTC Battery Removal	Setting	Clear BIOS Passwords on RTC Battery Removal has 2 options: • Disable • Enable When unchecked, the removal of the RTC battery WILL NOT clear the BIOS Administrator Password (BAP) and Power-on Password (POP). When checked, the removal of the RTC battery WILL clear the BIOS Administrator Password (BAP) and Power-on Password (POP). If Cover Removal Sensor is enabled and select to Administrator password or Administrator credential, then the Clear BIOS Passwords on RTC Battery Removal policy cannot be enabled and is also greyed out in F10. If the Clear BIOS Passwords on RTC Battery Removal policy is enabled, then Cover Removal Sensor Administrator password and Administrator credential cannot be enabled and is also greyed out in F10.	Unchecked	Desktop only, exclusive with Cover Removal Sensor administrator settings (Administrator password/Administrator credential) Reboot Required Checked = Enable in WMI
SSD Cover	Menu	Controls settings for SSD Cover Sensor.		Only on selected 2025 mobile workstations
Thunderbolt Option ROM	Setting	When checked, this setting enables the execution of UEFI Option ROM(s) on PCIe devices that are connected through a Thunderbolt adapter (i.e., PCIe devices behind the Thunderbolt adapter) during pre-boot.	Unchecked	When set to Enable, a local unauthenticated attacker with physical access to a user's system may be able to obtain read or write access to main memory via a DMA attack during platform boot. Checked = Enable in WMI
Hard Drive Utilities	Menu	Utilities to protect private information on individual hard drives: Drive Lock and Secure Erase.		
Absolute Persistence Module Current State	Label	A subscription service that provides PC theft recovery, tracking and data delete solutions.		
Activation Status	Display Only	The subscription status can be inactive, active, or permanently disabled.	Inactive	

Feature	Type	Description	Default	Notes
Absolute Persistence Module Permanent Disable	Display Only	Shows current state of the Absolute Persistence module (Yes = disabled, No = available).	No	
System Management Command	Setting	When checked, allows authorized HP service personnel in possession of the PC to reset security settings in case of a customer service event. For customers that require more BIOS security, uncheck this to prevent this type of HP service command. NOTE: If BIOS password is lost and this option is disabled, HP authorized personnel cannot remove a lost password.	Checked	Reboot Required Checked = Enable in WMI
Restore Security Settings to Factory Defaults	Action	Apply factory defaults to all security settings. NOTE: Escaping (ESC) at the Reset Request screen will leave settings as they were except for the Administrator & Power-on passwords which are still cleared.		Reboot Required

4.1 Password Policies Menu

This submenu allows the administrator to set text requirements controlling the use of symbols, numbers, case, and spaces for the BIOS administrator password and the power-on password. To access this menu, a password must be already set. Changes to these policies do not apply retroactively to existing passwords.

Table 11 Password Policies Menu features

Feature	Type	Description	Default	Notes
Password Minimum Length	Setting	Allows the administrator to specify the minimum number of characters required for a password. - Minimum: 4 - Maximum: 32	8	
At least one symbol required in Administrator and User passwords	Setting	When checked, passwords require at least one symbol, such as \$, %, ^, &, or #	Unchecked	WMI Name "At least one symbol is required in Administrator and User passwords" Checked = Enable in WMI

Feature	Type	Description	Default	Notes
At least one number required in Administrator and User passwords	Setting	When checked, passwords require at least one number	Unchecked	WMI Name "At least one number is required in Administrator and User passwords" Checked = Enable in WMI
At least one upper-case character required in Administrator and User passwords	Setting	When checked, passwords require at least one upper case character	Unchecked	WMI Name "At least one upper case character is required in Administrator and User passwords" Checked = Enable in WMI
At least one lower-case character required in Administrator and User passwords	Setting	When checked, passwords require at least one lowercase character	Unchecked	WMI Name "At least one lower case character is required in Administrator and User passwords"
Are spaces allowed in Admin and User passwords?	Setting	When checked, passwords can have one or more spaces	Unchecked	WMI Name "Are spaces allowed in Administrator and User passwords?" Checked = Enable in WMI
Allow User to Modify Power-on Password	Setting	Options are No, Change Only, and Change or Delete	Change or Delete	
Wake on LAN Power-on Password Policy	Setting	Options are Require Password and Bypass Password	Require Password	
Prompt for Power-on Password on OS Restart	Setting	If enabled, the system will prompt for Power-on Password on restart/reboot. If disabled, the system will bypass Power-on Password on restart/reboot. This setting does not impact prompts for Power-on Password on system power on or hibernate resume.	Checked	Checked = Enable in WMI

4.2 Administrator Authentication Policies Menu

This submenu allows the administrator to set limitations to some boot features, such as administrator permissions, requiring the user to enter an administrator password. To access this menu, a password or a Sure Admin Local Access Key must be already set.

Table 12 Password Policies Menu features

Feature	Type	Description	Default	Notes
Prompt for Admin authentication on F9 (Boot Menu)	Setting	When checked, the administrator password is required to enter the boot menu.	Unchecked	Old Name "Prompt for Admin password on F11 (System Recovery)"
Prompt for Admin authentication on F11 (System Recovery)	Setting	When checked, the administrator password is required to enter system recovery.	Unchecked	Checked = Enable in WMI Old Name "Prompt for Admin password on F11 (System Recovery)"
Prompt for Admin authentication on F12 (Network Boot)	Setting	When checked, the administrator password is required to enter the network boot menu.	Unchecked	Checked = Enable in WMI Old Name "Prompt for Admin password on F12 (Network Boot)"
Prompt for Admin authentication on Capsule Update	Setting	When checked, the administrator password is required to process a firmware capsule update.	Unchecked	Removed from 2022 and later platforms
BIOS Administrator visible at Power-on Authentication	Setting	When <i>not</i> checked, there is only a prompt for the Power-on password.	Checked	Checked = Enable in WMI

4.3 Trusted Platform Module (TPM) Embedded Security Menu

This submenu is for the Trusted Platform Module (TPM). TPM is a dedicated microprocessor that provides security functions for secure communication and software and hardware integrity. The built-in TPM hardware solution is more secure than a software-only solution.

Table 13 TPM Embedded Security Menu features

Feature	Type	Description	Default	Notes
TPM Specification Version	Display Only	The Trusted Computing Group (TCG) is an industry group that defines specifications for a TPM. As of this writing, possible TPM specification versions are 1.2 or 2.0. NOTE: Windows 10 and Windows 11 require TPM 2.0 capability.	2.0	

Feature	Type	Description	Default	Notes
TPM Device	Setting	Makes the TPM available. The following settings are possible: - Available - Hidden	Available	Reboot, Physical Presence Required
TPM State	Setting	When checked, enables the ability for the OS to take ownership of the TPM or Enables OS and application access to the various security capabilities of the TPM (v2.0).	Checked	Reboot, Physical Presence Required Checked = Enable in WMI This setting is for 2025 and earlier platforms only.
Clear TPM	Action	When selected, clears the TPM on the next boot. After clearing the TPM, this resets to No. The following settings are possible: - No - On next boot	No	Reboot Required
TPM Activation Policy	Setting	This setting allows an administrator to choose between convenience and extra security. The extra security is to ensure that the user of the system will at least see that the TPM device upgraded its firmware (F1 to Boot), or at most the user has the ability to reject the upgrade of the TPM device (Allow user to reject). These user prompts limit the impact of remote attacks on the system by requiring a user to be physically present for the upgrade. When security of the system is of less concern, the third option (No prompts) removes any requirement for a user to acknowledge the upgrade. This last option is the most convenient for remotely upgrading many systems at once. The following settings are possible: - F1 to Boot - Allow user to reject - No prompts	Allow user to reject	HP recommends an option that requires the physical presence of the user
Pluton Security Processor	Setting	When the Pluton Security Processor is active, additional security features defined by Microsoft are enabled – including compliance with Microsoft Secure Core PC (SCPC) criteria.	Enabled on AMD Cannot be disabled on Intel	Setting available only on HP G1a, G1i Products

4.4 BIOS Sure Start Menu

Settings menu for enhanced hardware-based assurance that only HP approved Embedded Controller firmware will run on the HP Embedded Controller and that only HP approved BIOS will run on the host CPU.

Table 14 BIOS Sure Start Menu features

Feature	Type	Description	Default	Notes
Verify Boot Block on every boot	Setting	When not checked, HP Sure Start verifies the integrity of HP firmware in the nonvolatile (flash) memory before resume from Sleep, Hibernate, or Off. When checked, HP Sure Start verifies the integrity of HP firmware in the nonvolatile (flash) memory across operating system restart (warm reset) in addition to resume from Sleep, Hibernate Off. This setting provides higher security assurance but could increase the time required to restart the operating system.	Unchecked	Reboot Required Checked = Enable in WMI
Network Controller Configuration Restore	Action	Network Controller Configuration Restore This action restores the network controller parameters to the factory state saved in the HP Sure Start Private nonvolatile (flash) memory. NOTE: This process can take up to 30 seconds. You need to restore this only when the Network Controller Configuration mismatch warning is set.		Reboot Required
Prompt on Network Controller Configuration Change	Setting	When checked, HP Sure Start will monitor the network controller configuration and prompt the local user if any changes are detected compared to the factory configuration. The local user has the option to ignore the prompt or restore the network controller to the factory configuration when prompted.	Checked	Intel Only Reboot Physical Presence Required Checked = Enable in WMI
Dynamic Runtime Scanning of Boot Block	Setting	When checked, allows HP Sure Start to verify the integrity of the HP firmware in the nonvolatile (flash) memory every 15 minutes while the system is on and the operating system is running. NOTE: Available on both NB and DT Intel platforms.	Checked	Intel only Checked = Enable in WMI

Feature	Type	Description	Default	Notes
Sure Start BIOS Settings Protection	Setting	Protects critical BIOS Settings by saving a backup copy and restoring them if altered.	Unchecked	Not accessible with no Admin credentials set Checked = Enable in WMI
Sure Start Secure Boot Keys Protection	Setting	Saves backup copy of Secure Boot Keys so that they can be recovered if someone attempts to alter them in an unauthorized manner.	Checked	Reboot Required
Enhanced HP Firmware Runtime Intrusion Prevention and Detection	Setting	Monitors key areas of memory for corruption or attack, notifies user of attack (based on the settings in Sure Start Security Event Policy), and prevents the attack from taking place. NOTE: Available on Intel Sure Start platforms that support this feature. Available on AMD Sure Start platforms 2020 and later. NOTE: In 2026 and later platforms this setting was removed from F10 but the WMI setting remains and is read-only (cannot be disabled)	Checked	Checked = Enable in WMI
Sure Start Security Event Policy	Setting	Determines how to respond to a detected event Sure Start security event (BIOS recovery, Memory Intrusion, etc.): - Log event only - Log the event in the audit log. - Log Event and notify user - Log the event in the audit log and prompt the user to acknowledge the event. - Log Event and power off system - Log the event in the audit log and power off the system. NOTE: Not available prior to 2016.	<2026 = Log Event and notify user >2027 = Log event only	Logs will be captured and available to IT Admins, regardless of Default setting.
Sure Start Security Event Boot Notification	Setting	When notifying the user of a detected Sure Start security event during boot, this determines whether the message should: - Time out after 15 seconds - Require Acknowledgment	Require Acknowledgment	

Feature	Type	Description	Default	Notes
Virtualization Based BIOS Protection	Setting	Uses virtualization hardware to protect HP BIOS from UEFI Expansion driver and PCI Expansion ROM driver modules that attempt to access BIOS resources and hardware configuration. This protection is put in place during early boot of BIOS to protect BIOS code/critical platform resources from these drivers and unloads when BIOS passes control to OS. NOTE: On Intel platforms, enabling this setting requires Virtualization Technology (VTx) to be enabled. On AMD platforms, enabling this setting requires SVM CPU Virtualization to be enabled. NOTE: Beginning in 2024, this setting is only available on PCs with Enterprise Security Edition.	Checked	Reboot required Checked = Enable in WMI
Virtualization Based BIOS Protection Manual Recovery	Setting	This setting determines the behavior of the system when an issue is detected by the Virtualization Based BIOS Protection feature. When unchecked, the system will restart and boot with limited functionality to display an error message to the user and query the user for the action to take. When checked, the system will halt allowing the administrator to perform forensics. When the manual recovery sequence is provided by the user, the system will follow the same behavior as when unchecked. NOTE: Beginning in 2024, this setting is only available on PCs with Enterprise Security Edition.	Unchecked	Reboot required Checked = Enable in WMI

4.5 Secure Boot Configuration Menu

This sub menu is for the Secure Boot OS loader feature.

Table 15 Secure Boot Menu features

Feature	Type	Description	Default	Notes
Secure Boot	Setting	When checked, enables the Secure Boot capability.	Checked	Enabled on Windows systems and disabled on non-Windows (FreeDos, Ubuntu)?

Platform Key	Display only	- HP - Not Enrolled - Custom	HP	Factory will ship with 'HP' keys WMI Name: "Secure Boot Platform Key"
Import Custom Secure Boot keys	Setting	When checked and system is rebooted, custom secure boot keys are imported from the EFI\HP directory from the hard drive or USB device. The custom keys consist of PK, KEK, DB, and Dbx .bin files. When import succeeds or fails, a preboot prompt shows the results of each key bin file. NOTE: This setting cannot be changed (i.e., will be grayed out) when "Sure Start Secure Boot Keys Protection" is enabled.	Unchecked	Reboot Required Checked = Enable in WMI
Clear Secure Boot keys	Action	When checked, clears the Secure Boot keys one time on next save and exit. This setting will be unchecked again when you return from exit. This action is not available when no imported keys are present. NOTE: This setting cannot be changed (i.e., will be grayed out) when "Sure Start Secure Boot Keys Protection" is enabled.	Unchecked	Reboot Required Checked = Enable in WMI
Reset Secure Boot keys to factory defaults	Action	When checked, restores secure boot keys to factory defaults one time on next save and exit. This setting will be unchecked again when you return from exit. NOTE: This setting cannot be changed (i.e., will be grayed out) when "Sure Start Secure Boot Keys Protection" is enabled.	Unchecked	Reboot Required Checked = Enable in WMI
Windows UEFI CA 2023	Setting	Enable this setting when using Windows 2023-certified bootloader files. When enabled, it includes the Windows UEFI CA 2023 certificate in the active and default UEFI DB. Caution: Before changing this BIOS setting, BitLocker should be suspended to avoid triggering BitLocker recovery.	Checked	Reboot Required upon setting change This setting contains a policy ownership state to track whether it has been modified by the user. If this setting is explicitly changed, the BIOS takes control of the contents in the UEFI DB. Checked = Enable in WMI
Microsoft Option ROM UEFI CA 2023	Setting	Enable this setting when using a third-party Option ROM. When enabled, it includes the Microsoft Option ROM CA 2023 certificate in the active and default UEFI DB. (Option ROMs are commonly used with graphics, networking, and storage controller cards.)	Unchecked	Reboot required upon setting change. To Improve Security HP recommends keeping this setting disabled (unchecked) when not using a third-party Option ROM.

		Caution: Before changing this BIOS setting, BitLocker should be suspended to avoid not triggering BitLocker recovery.		This setting contains a policy ownership state to track whether it has been modified by the user. If this setting is explicitly changed, the BIOS takes control of the contents in the UEFI DB. Checked = Enable in WMI
Microsoft UEFI CA 2023	Setting	Enable this setting when using third-party bootloaders or third-party EFI applications. When enabled, it includes the Microsoft UEFI CA 2023 certificate in the active and default UEFI DB. Caution: Before changing this BIOS setting, BitLocker should be suspended to avoid triggering BitLocker recovery.	Unchecked	Reboot required upon setting change To improve security, HP recommends keeping this setting disabled (unchecked) when not using third-party boot loaders or EFI applications. This setting contains a policy ownership state to track whether it has been modified by the user. If this setting is explicitly changed, the BIOS takes control of the contents in the UEFI DB.. Checked = Enable in WMI
Enable MS UEFI CA key	Setting	Enable this setting when using third-party bootloaders, third-party EFI applications, or third-party option ROMs. When enabled, the BIOS includes the Microsoft UEFI CA 2011 certificate in the active and default UEFI DB. Caution: Before changing this BIOS setting, BitLocker should be suspended to avoid triggering BitLocker recovery.	For Windows OS: Checked / Unchecked (See Notes) Android: Unchecked Otherwise: Checked	Reboot required upon setting change To improve security, HP recommends this setting be disabled (unchecked) for systems running Windows OS when not using third-party bootloaders, third-party EFI applications, or third-party option ROMs signed with a Microsoft UEFI CA 2011 signature. The latest HP factory preinstall process unchecks this BIOS setting unless one of the following is included in the system configuration: 1) Non-Windows operating system 2) Option ROM support is not included in the system BIOS. 3) "Microsoft 3rd Party UEFI CA Enable" AV part number is

				included. This AV part number is intended for customers who have Secure Boot enabled when using third party EFI code like full volume encryption software, third party image recovery software, or third-party option ROMs included with add-in graphics and networking cards. Checked = Yes in WMI
Ready BIOS for Device Guard Use	Action	Ready BIOS for Device Guard Use includes a drop-down box that automatically configures the BIOS settings that Windows requires to enable Device Guard or to change the configuration back to the configuration before Device Guard was enabled. Device Guard is a Windows feature that enables higher security around drivers and BIOS behavior. The following settings are possible: • Do Nothing • Configure on Next Boot • Clear Configuration on Next Boot When set to Configure on Next Boot, the BIOS changes the following settings to the states required by Device Guard after saving changes and exit. • Virtualization features are enabled. • Removable and network boot devices are disabled (for example, USB boot, CD-ROM boot, Thunderbolt™ boot, etc.). • MS UEFI CA Key is disabled. When set to Clear Configuration on Next Boot, the BIOS sets the listed features to their Custom Default state if custom defaults have been saved. If custom defaults have not been saved, the BIOS restores the listed features to their factory default states.		This setting is removed on 2026 and later platforms

4.6 Secure Platform Management (SPM)

This submenu controls settings for Secure Platform Management that are used for secure enablement and management of the HP Sure Run, Sure Recover, and Sure Admin (Enhanced BIOS Authentication Mode) capabilities.

You cannot provision SPM and activate HP Sure Run directly from the BIOS Setup interface. You can provision SPM using HP Client Security Manager Software or the HP Manageability Integration Kit. When provisioned, the controls in this menu can be used to deprovision the system or deactivate HP Sure Run.

Table 16 Secure Platform Management Menu features

Feature	Type	Description	Default	Notes
HP Cloud Managed	Label			
HP Cloud Managed State	Display Only	This display-only field shows the state of the HP Cloud Managed support. • Enabled • Disabled • Permanently Disabled If permanently disabled, the feature cannot be used and requires HP Service to re-enable it. (This setting was introduced in the second half of 2021)		Default to the value by HP Cloud Managed setting
HP Cloud Managed	Setting	When checked, this setting allows customers to use HP cloud based manageability tools (e.g., Wolf Security Controller & TechPulse/WXP) to provision / rotate / remove the cryptographic keys used with features such as Wolf Protect & Trace, Sure Admin and Sure Recover, in a simple straightforward manner. If the owner of the machine does not require this capability or would prefer that HP tools not be able to do this, they can use this setting to disable this feature. • (This setting was introduced in the second half of 2021) • Access to this setting is disabled when threshold key is set	Checked	Checked = Enable in WMI
Display Enrolled Management Keys	Menu	• Shows table with Key ID number and owner name associated with the management key		Shows User: Key Owner List
Remote Device Management	Label			
Remote Device Management Status	Display Only	This display-only field shows the state of Remote Device Management support. • Enabled • Disabled • Permanently Disabled If permanently disabled, the feature cannot be used and requires an HP Service to re-enable. (This setting was introduced in the second half of 2021)		Default to the value by Remote Device Management setting
Remote Device Management	Setting	When checked, this setting allows a highly secure set of remote management operations (e.g. lock and wipe) to be performed on the machine.	Checked	Checked = Enable in WMI

		This setting cannot be disabled while Wolf Protect & Trace is enabled. (This setting was introduced in the second half of 2021)		
SPM	Label			
SPM Current State	Display Only	- Provisioned - Not provisioned	Not provisioned	WMI Name "Secure Platform Management Current State"
Unprovision SPM	Action	This action deprovisions SPM, which causes HP Sure Run to revert to the Inactive state, and returns HP Sure Recover to default settings.		Not available via WMI
HP Sure Admin EBAM BIOS Authentication Mode (EBAM)	Label			
HP Sure Admin – EBAM Current State	Display Only	This displays the current state of Sure Admin's Enhanced BIOS Authentication Mode which controls whether Sure Admin authorization strings or BIOS Admin Password must be used when changing BIOS settings via WMI. Changing this value must be done using WMI (see notes column). - Enabled - Disabled	Disabled	WMI Name "Enhanced BIOS Authentication Mode" The values used in WMI are Enable and Disable. When changing this setting an authorization string is ALWAYS required.
Disable EBAM	Action	This action disables Enhanced BIOS Authentication Mode (EBAM)		Not available via WMI
Local Access Key	Display Only	Indicates whether there is a Local Access Key configured in the system. - Not Present - Present		Not available via WMI
Clear EBAM Local Access Key(s) and Reboot	Action	This action deletes all currently established local access keys created for Enhanced BIOS Authentication Mode (EBAM)		Not available via WMI
System Locking	Setting (Display Only)	- Unavailable - Requires Local Access Key to be present. - Available from <ESC> Startup Menu.		System with On Demand Lock feature
HP Sure Run	Label			

HP Sure Run Current State	Display Only	- Active - Inactive	Inactive	
Deactivate HP Sure Run	Action	This action deactivates HP Sure Run without deprovisioning SPM.		Not available via WMI

4.7 Smart Cover Menu

This submenu controls settings for Cover Lock and Cover Sensor Including those associated with the HP Tamper Lock feature. These settings are only present when the implied optional devices are installed in the system.

Table 17 Smart Cover Menu features

Feature	Type	Description	Default	Notes
Cover Lock	Setting	The Smart Cover Lock is a software-controllable solenoid lock. This lock restricts unauthorized access to the system's internal components. The following settings are possible: - Lock - Unlock	Unlock	Desktop with Cover Lock Reboot Required
Cover Removal Sensor	Setting	The Cover Removal Sensor has the following settings: Disable Notify the User: Displays warning message on next boot if opened. Administrator Password (when password is set or Sure Admin Enhanced BIOS Authentication Mode is enabled with a Local Access Key set): Requires entering the administrator password or the PIN (if Local Access Key is present) before continuing to boot after the cover is opened. Administrator Credential: exactly the same behavior as Administrator Password.	Disable	System with Cover Sensor Reboot Required Administrator Credential may not be available on all systems that support Smart Cover.
Power off upon cover removal	Setting	When checked, if the cover is removed while the system is on or asleep (S3 or Modern Standby), then the system will immediately power down. This setting is only active and can only be modified while Cover Sensor Removal is enabled. This only affects cover removals that occur after the setting is set.	Unchecked	May not be available on all systems that support Smart Cover. WMI Name: "Power Off Upon Cover Removal" Checked = Enable in WMI
Clear TPM on boot after cover removal	Setting	When checked, if the cover is removed, then TPM will be cleared on the boot after the cover was removed. This setting is only active and can only be modified while Cover Removal Sensor is enabled. These only affects cover removals that occur after the setting is set.	Unchecked	May not be available on all systems that support Smart Cover. Checked = Enable in WMI

Last Cover Removal and Count	Display Only	This reports the last time the cover was removed and how many times it has been removed and acknowledged since it left the factory, in the following format: MM/DD/YYYY HH:MM:SS. X times. Depending on system factors, consecutive cover removals may count as a single cover removal. The date and time may be reported as all 0's in cases where the value cannot be determined such as real-time clock power loss.	0 times	May not be available on all systems that support Smart Cover.
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4.8 Hard Drive Utilities Menu

This submenu provides features that protect the data on individual hard drives, such as recovering the master boot record (MBR), preventing unauthorized access, and erasing data.

Table 18 Hard Drive Utilities Menu features

(Next Page)

Feature	Type	Description	Default	Notes
Save/Restore GPT of System Hard Drive	Setting	When checked, saves a baseline GUID Partition Table that can be restored if a change is detected.	Unchecked	Reboot Required WMI Name "Save/Restore GPT of System Hard Drive" Checked = Enable in WMI
Boot Sector (GPT) Recovery Policy	Setting	Allows selection of the default action when an MBR/GPT event occurs. - Local user control - Recover in event of corruption	Local User Control	WMI Name "Boot Sector (MBR/GPT) Recovery Policy"
DriveLock/Automatic DriveLock	Menu	DriveLock prevents unauthorized access to the contents of a selected hard drive.		
Prompt for DriveLock Password on OS Restart	Setting	When checked, BIOS will prompt for a DriveLock Password on restarts. When unchecked, BIOS will automatically unlock drives on restarts. NOTE: When the setting is unchecked, all drives must be configured to use the same password.	Checked	Checked = Enable in WMI Old Name "Drivelock Password on restart"

Secure Erase (Select a Drive...)	Action	Uses hardware-based methods safely to erase all data and personal information from a selected Hard Drive. Prior to Q3-2024, HP Secure Erase was only available on HP DTWKS platforms. In September 2024, HP Secure Erase was backported and distributed on ≥G9 HP PCs with HP Commercial BIOS (Applied to ~240 DT and NB Platforms from 2022 or later). In 2026, HP Secure Erase added user options to select Purge or Clear, and NIST compliance (with verification) or speed (without verification). HP Secure Erase with verification is compliant with NIST 800-88 Rev. 1 Guidelines for Media Sanitization Secure Erase will default to the highest level of sanitization the drive is capable of (Level 1 Clear or Level 2 Purge) At conclusion of the drive sanitization, the PC presents a document of completion that can be saved to a USB drive. If the Secure Erase method is NIST compliant, the resulting document includes the following items in the NIST 800-88r1 Document of Drive Sanitization. ▪ Date of Drive Sanitization ▪ Time of Drive Sanitization ▪ Drive Manufacturer, Type, Serial Number ▪ Sanitization Type ▪ HP Secure Erase Version (BIOS) ▪ Drive Hash Data (Before/After)	Local User Control	Reboot Required Not available via WMI
Allow OPAL Hard Drive SID Authentication	Setting	BIOS supports drive encryption using DriveLock feature by creating the storage device's ownership key. If BIOS creates the key, any 3rd party applications (including other encryption software) are not allowed to perform certain drive operations such as establishing their own key using SID. Encryption software applications may or may not be limited by SID authentication lockout depending on how they are designed.	Unchecked	Reboot Required Checked = Enable in WMI
Restore NVMe Namespace configuration (Restore Low Level Drive Layout)	Action	Selecting this option will delete the existing NVMe namespaces and apply the system's previously saved configuration.		

4.9 DriveLock/Automatic DriveLock Menu

DriveLock prevents unauthorized access to the contents of a selected hard drive. Enter a password to access the drive and the drive is accessible only when attached to a PC.

NOTE: DriveLock states cannot change after a warm reboot for SATA drives. Power off the system and then boot directly to the BIOS setup to access these menus. The DriveLock Master/Administrator and User passwords cannot be changed if you enable Automatic DriveLock.

Table 19 DriveLock Menu features

Feature	Type	Description	Default	Notes
Automatic DriveLock	Setting	DriveLock is intended to prevent someone from accessing data on your drive after they have physically removed it from your system. A BIOS administrator password is required for this feature. When this setting is checked, the BIOS sets a randomly generated user password, sets the master password with the BIOS administrator password, and marks the drive as a member of an Automatic DriveLock group. Thereafter, the BIOS automatically unlocks the drive while it is attached to its host system. If the drive is physically removed from its host system and attached to another system, the user is prompted for the DriveLock password. The user must provide the BIOS administrator password from the original host system to access the drive.	Unchecked	Power cycle required Not in WMI
Disable DriveLock	Action	Displayed only if DriveLock is enabled and a valid password was supplied at the DriveLock POST prompt. Allows DriveLock to be disabled when it is enabled.		Power cycle required Not in WMI
Set DriveLock Master Password Set DriveLock User Password	Setting	Creates another password to access a hard drive with DriveLock protection.		Power cycle required Not in WMI
Enable DriveLock	Action	Enables DriveLock protection and creates a user password distinct from the master password that allows access to the hard drive (SATA drives). For NVMe type drives in the M.2 slot, this requires setting an administrator password instead of a user password.	Disable	Power cycle required Not in WMI
Change DriveLock User Password	Action	Displayed only if DriveLock is enabled and a valid password was supplied at the DriveLock POST prompt. Allows the user password to be changed when selected.		Power cycle required Not in WMI

Feature	Type	Description	Default	Notes
Automatic DriveLock	Setting	DriveLock is intended to prevent someone from accessing data on your drive after they have physically removed it from your system. A BIOS administrator password is required for this feature. When this setting is checked, the BIOS sets a randomly generated user password, sets the master password with the BIOS administrator password, and marks the drive as a member of an Automatic DriveLock group. Thereafter, the BIOS automatically unlocks the drive while it is attached to its host system. If the drive is physically removed from its host system and attached to another system, the user is prompted for the DriveLock password. The user must provide the BIOS administrator password from the original host system to access the drive.	Unchecked	Power cycle required Not in WMI
Disable DriveLock	Action	Displayed only if DriveLock is enabled and a valid password was supplied at the DriveLock POST prompt. Allows DriveLock to be disabled when it is enabled.		Power cycle required Not in WMI
Change DriveLock Master/Administrator Password	Action	Displayed only if DriveLock is enabled and a valid password was supplied at the DriveLock POST prompt. Allows the master (SATA) or administrator (NVMe) password to be changed when selected.		Power cycle required Not in WMI

4.10 SSD Cover Menu

This submenu controls settings for SSD Cover Sensor Including those associated with the SSD Cover Sensor removal feature. These settings are only present when the implied optional devices are installed in the system.

Table 20 SSD Cover Menu features

Feature	Type	Description	Default	Notes
SSD Cover Removal Sensor	Setting	The SSD Cover Removal Sensor has the following settings: • Disable • Notify User: Displays warning message on next boot if opened.	Disable	System with SSD Cover Sensor Reboot Required
Power off upon SSD cover removal	Setting	When checked, if the SSD cover is removed while the system is on or asleep (S3 or Modern Standby), then the system will immediately power down. This setting is only active and can only be modified while SSD Cover Sensor Removal is enabled. This only affects SSD cover removals that occur after the setting is set.	Unchecked	Checked = Enable in WMI

5 Advanced Menu

MainSecurityAdvancedUEFI Drivers



HP Computer Setup

- ⇒

Display Language
- ⇒

Scheduled Power-On
- ⇒

Boot Options
- ⇒

HP Sure Recover
- ⇒

System Options
- ⇒

Built-In Device Options
- ⇒

Port Options
- ⇒

Network Settings
- ⇒

Power Management Options
- ⇒

Remote Management Options (Intel Only)
- ⇒

Electronic Labels (Notebook & AiO Only)
- ⇒

MAC Address Pass Through (Notebook Only)
- ⇒

Thunderbolt™ Options (2019+ with TBT)
- Remote HP PC Hardware Diagnostics
- ⇒

Settings
- ⇒

Execute Remote HP PC Hardware Diagnostics

5.1 Advanced Menu

For detailed information on the features in the advanced menu, see the following table:

Table 21 Advanced Menu features

Feature	Type	Description	Default	Notes
Display Language	Menu	Select the display language and the keyboard language. Choose between 15 languages. You can display the menu in English, French, German, Spanish, Italian, Dutch, Danish, Japanese, Norwegian, Portuguese, Swedish, Finnish, Simplified Chinese, Traditional Chinese, or Russian. NOTE: Affects the BIOS menus, not the OS nor the WMI commands. Russian language support is only available in the most recent product generations.		
Keyboard Part Number Suffix After Replacement	Menu	HP provides a BIOS option that allows ITDM end users to select the correct keyboard language layout when it differs from the original factory configuration, ensuring it matches the newly installed keyboard. After installing a new keyboard, ITDM end users will need to go to BIOS F10 menu to update the layout setting. The BIOS presents the following Help Text: <i>? This setting should be used only when replacing a keyboard on the PC with new keyboard that has a different language layout.</i> The BIOS will display input fields for a range of 3-4 characters, from the model of the keyboard displayed on the shipping box the keyboard was packaged in. E.g., JPN2, KOR, BRZL, GR, TURK-F, OTHER NOTE: If the F10 Restore Factory Defaults option is used, the PC will revert to the original keyboard layout (as shipped from the factory, determined by a Feature Byte) and the BIOS setting will need to be reset to align with keyboard-language configuration that is in the PC.		F10 only. Not available as a WMI, BCU or CMSL controlled setting.
Scheduled Power On	Menu	Choose days of the week and a single time of day for the system to turn on. This feature wakes the system up from a turned-off state.		
Boot Options	Menu	Settings that control the behavior of the system during boot up.		
HP Sure Recover	Menu	Settings that control when and how the BIOS should attempt to reinstall the operating system. Also called <i>OS Recovery</i> .		
System Options	Menu	Settings that control the CPU, PCI, PCIe, the power button, and function keys.		
Built-In Device Options	Menu	Settings of other devices built into the PC.		
Port Options	Menu	Settings that enable or disable ports and interrupts on the system.		
Network Settings	Menu	Various options for Network Settings.		
Power Management Options	Menu	Settings that control power saving features and the behavior of the system in low power modes.		

Feature	Type	Description	Default	Notes
Remote Management Options	Menu	Settings that control Intel Active Management technology that provides out-of-band remote management of the system.		Intel Only
Electronic Labels	Display Only	Mandatory certification marks, for example the Federal Communication Commission (FCC) Declaration of Conformity (Doc) and the CE marking for Europe.		Notebook and All-in-One Only
MAC Address Pass Through	Menu	Configure a custom Host Based MAC Address (HBMA) for the system as well as define the priority of Network Interface Cards (NIC).		Notebook Only
Thunderbolt Options	Menu	Configure options related to Thunderbolt		
Remote HP PC Hardware diagnostics	Label	Remote HP PC Hardware diagnostics.		
Settings	Menu	Settings for Remote HP PC Hardware diagnostics.		
Execute Remote HP PC Hardware Diagnostics	Action	When selected, will download and run HP Remote Diagnostics.		

5.2 Display Language Menu

This submenu allows for selection of the display language and the keyboard language. For each setting, choose from the following languages:

- English
- Deutsch
- Español
- Italiano
- Français
- 日本語
- Português
- Dansk
- Svenska
- Nederlands
- Norsk
- Suomi
- 简体中文
- 繁體中文
- русский

NOTE: Affects the BIOS menus, not the OS nor the WMI commands.

Table 22 Display Language Menu features

Feature	Type	Description	Default	Notes
Select Language	Setting	Language used by BIOS setup menus.	English	
Select Keyboard Layout	Setting	Language of the keyboard layout used by BIOS setup menus.	English	

5.3 Scheduled Power-On Menu

This submenu controls the days of the week and a single time of day for the system to turn on the computer. This feature wakes the system up from a powered-off state.

Table 23 Scheduled Power-On Menu features

Feature	Type	Description	Default	Notes
Sunday Monday Tuesday Wednesday Thursday Friday Saturday	Setting	Days of the week selection.		Reboot Required
Hour	Setting	Time selection.	0	Reboot Required WMI Name "BIOS Power-On Hour"
Minute	Setting	Hour: 0 – 23, Minute: 0 – 59.	0	Reboot Required WMI Name "BIOS Power-On Minute"

5.4 Boot Options Menu

This submenu controls the behavior of the system during boot up.

Table 24 Boot Options Menu features

Feature	Type	Description	Default	Notes
Startup Delay (sec.)	Setting	Select the number of seconds (0 – 60) to pause the boot before starting the OS. Increasing the delay gives more time to press a key that accesses one of the startup options, such as BIOS Setup (F10).	0	
Fast Boot	Setting	When checked, reduces boot up time by bypassing boot to USB, CD-ROM, and PXE. This skips some preboot initialization steps. NOTE: When a power-on password, other security features, or current boot order have been modified, Fast Boot is ignored.	Checked	Checked = Enable in WMI

Feature	Type	Description	Default	Notes
CD-ROM Boot	Setting	When checked, allows the system to boot from CD-ROM.	Checked	Checked = Enable in WMI
USB Storage Boot	Setting	When checked, allows the system to boot from USB devices.	Checked	Checked = Enable in WMI
Network (PXE) Boot	Setting	When checked, allows the system to boot from a network card if it supports PXE or UEFI network boot capability.	Checked	Checked = Enable in WMI
After Power Loss	Setting	Specifies the desktop state after power loss. The following settings are possible: - Power Off - Power On Previous State	Power Off	Desktop Only
IPv6 during UEFI Boot	Setting	When checked, allows the system to process IPv6 packets in pre-boot.	Checked	Checked = Enable in WMI
Network Boot TFTP Window Size	Setting	Select the TFTP Window Size (1-65535) used during a Network Boot. The Window Size refers to the number of consecutive blocks transmitted before stopping and waiting for the reception of the acknowledgment of the last block transmitted.	4	The default value is set by EDK2.
Power On When AC Detected	Setting	When checked, the notebook will turn on when it is off, when AC power has not been available and then becomes available.	Unchecked	Notebook Only Checked = Enable in WMI
Power On When Lid is Opened	Setting	When checked, the system turns on when the lid opens.	Unchecked	Notebook Only Checked = Enable in WMI
Prompt on Battery Errors	Setting	When checked, the system pauses during system boot to warn about battery errors.	Checked	Notebook Only Checked = Enable in WMI
Prompt on Memory Size Change	Setting	When checked, notify the user during the boot process when a memory size change has been detected. NOTE: This setting will be replaced by a dropdown if the system supports locked on memory size change. The system will be locked when a memory size change has been detected, and only administrator can unlock the system if this setting is set to Administrator Credential. - Disable - Enable - Administrator Credential	Checked	The default is Enable if the system supports locked on memory size change

Feature	Type	Description	Default	Notes
Prompt on Fixed Storage Change	Setting	When checked, notify the user during the boot process when a fixed storage change has been detected. NOTE: This feature will not report a change within a RAID configuration. NOTE: This setting will be replaced by a dropdown if the system supports locked on fixed storage change. The system will be locked when a fixed storage change has been detected, and only administrator can unlock the system if this setting is set to Administrator Credential. • Disable • Enable • Administrator Credential	Unchecked	The default is Enable if the system supports locked on fixed storage change
Audio Alerts During Boot	Setting	When checked, errors trigger audible beeps during POST.	Checked	Checked = Enable in WMI
NumLock on at boot	Setting	Set the keyboard Num Lock control to be on or off when system is booted.	Unchecked	WMI Name "NumLock on at boot" Checked = Enable in WMI
UEFI Boot Order	Action	Arrange the boot order from the UEFI devices found. By default, the system will arrange the boot order by device type using the following precedence: 1. USB 2. SATA DVD (Desktop Only) 3. SATA Hard Drives 4. M.2 devices 5. Network Boot Highlight the list and press Enter to adjust the order of the boot entries. If a new bootable device is added to the system, it appears at the bottom of the list, unless it is a USB device that uses the order of the USB placeholder already in the list.		

5.5 HP Sure Recover

Table 25 HP Sure Recover

Feature	Type	Description	Default	Notes
HP Sure Recover	Setting	If checked and HP Sure Recover is launched, the system firmware honors local and remote requests to reinstall the OS. If unchecked, Sure Recover is disabled and the operating system's default method is used.	Checked	WMI Name "OS Recover" Checked = Enable in WMI
Recover from Network	Setting	If checked, the system firmware checks the network repository for an updated recovery agent and may recover from the network. Otherwise, updates and recovery from the network is disabled.	Checked	Assuming Windows 10 or 11 is preinstalled. Gray when HP

Feature	Type	Description	Default	Notes
				Sure Recover is unchecked WMI Name "Recover OS from Network" Checked = Enable in WMI
Recover after Boot Failure	Setting	If this setting is enabled and no bootable UEFI OS is found, the system firmware launches HP Sure Recover.	Unchecked	Gray when HP Sure Recover is unchecked WMI Name "Recover OS after Boot Failure" Checked = Enable in WMI
Allow users to select Windows Recovery Options in F11	Setting	If this setting is enabled then when F11 is initiated there will be a button labeled "Launch WinRe" that will launch the Windows Recovery Menu	Unchecked	Gray when HP Sure Recover is unchecked
Prompt before Boot Failure Recovery	Setting	If this setting is enabled and HP Sure Recover is launched because of a boot failure, the user is notified of the boot failure and asked to choose whether to start or cancel HP Sure Recover.	Checked	Not shown if Recover after Boot Failure is unchecked Checked = Enable in WMI
Recovery Agent	Label			
URL:	Display Only	Location of the current recovery agent URL.	https://ftp.hp.com/pub/pcbios/CPR	Not shown unless Recover from Network checked WMI Name "OS Recovery Agent URL"
Username:	Display Only	Username (optional) to access the recovery agent.		Not shown unless Recover from Network checked WMI Name "OS Recovery Agent Username"
Provisioning Version:	Display Only	Version of the recovery agent's provisioning data. This value will be 0 until a scheduled download	0	Not shown unless Recover from Network checked

Feature	Type	Description	Default	Notes
		occurs after a change is made to the recovery agent URL.		WMI Name "OS Recovery Agent Provisioning Version"
OS Recovery Agent Version	Display Only	Version of the recovery agent stored in the embedded secure storage device. The Version will be updated when embedded secure storage device is updated.		Not shown unless an embedded secure storage device is installed. WMI Name "OS Recovery Agent Version"
Recovery Image	Label			
URL:	Display Only	Location of the current recovery image URL.		Not shown unless Recover from Network checked WMI Name "OS Recovery Image URL"
Username:	Display Only	Username (optional) to access the recovery image.		Not shown unless Recover from Network checked WMI Name "OS Recovery Image Username"
Provisioning Version:	Display Only	Version of the recovery image's provisioning data. This value will be 0 until a scheduled download occurs after a change is made to the recovery image URL.		Not shown unless Recover from Network checked WMI Name "OS Recovery Image Provisioning Version"
OS Recovery Image Version	Display Only	Version of the recovery image stored in the embedded secure storage device. The Version will be displayed when first time do an update on eMMC.		Not shown unless an embedded secure storage device is installed.
OS Recovery Driver Version	Display Only	Version of the recovery driver stored in the embedded secure storage device.		Not shown unless an embedded secure storage device is installed.
Recovery Image	Menu	Shows User: Failover URLs, Usernames, and Provisioning versions		

Feature	Type	Description	Default	Notes
Failover Repositories				
Failover Repository 1	Label			Not shown unless Recover from Network checked and Failover Repository is populated
URL:	Display Only	Location of the Failover 1 recovery image URL.		Not shown unless Recover from Network checked and Failover Repository 1 is populated WMI Name "OS Recovery Image Failover URL 1"
Username:	Display Only	Username (optional) to access the Failover 1 recovery image.		Not shown unless Recover from Network checked and Failover Repository 1 is populated WMI Name "OS Recovery Image Failover Username 1"
Provisioning Version:	Display Only	Version of the Failover 1 recovery image's provisioning data. This value will be 0 until a scheduled download occurs after a change is made to the recovery image URL.		Not shown unless Recover from Network checked and Failover Repository 1 is populated WMI Name "OS Recovery Image Provisioning Version 1"
Embedded Storage for Recovery	Display Only	Reports the size of the embedded secure storage device.	32 GB	Not shown unless an embedded secure storage device is installed.

5.6 System Options Menu

Table 26 System Options Menu features

Feature	Type	Description	Default	Notes
Configure Storage Controller for RAID	Setting	When checked, configures SATA Controller for RAID mode.	Unchecked	Select products only Checked = Enable in WMI
POST prompt for RAID Configuration	Setting	When checked, prompts user to hit a key for RAID configuration.	Unchecked	Select products only
Configure Storage Controller for Intel Optane	Setting	Enables driver support for NVMe Intel® Optane® storage module. Requires additional configuration by Intel Rapid Storage Technology software application. IMPORTANT: After Optane is initialized in the OS, do not boot with this setting disabled. The OS may become corrupted unless Optane is unconfigured first.	Unchecked	Intel Only Checked = Enable in WMI
Configure Storage Controller for VMD	Setting	When checked, the Intel Volume Management Device (VMD) controls the storage devices in the system, allowing support of RAID and Optane features.	Depends on factory storage configuration	Select products only Checked = Enable in WMI
Limit PCIe Speed	Setting	Allows you to restrict the maximum speed of the PCI Express devices to previous generations. The following settings are possible: • Auto • Gen 1 (2.5Gbps) • Gen 2 (5Gbps) • Gen 3 (8Gbps)	Auto	Desktop Workstations Only
Turbo Boost	Setting	When checked, enables Intel Turbo Boost Technology to improve performance when operation conditions allow.	Checked	Intel Only WMI Name "Turbo-boost" Checked = Enable in WMI

Feature	Type	Description	Default	Notes
Hyper-threading (Intel HT)	Setting	When checked, enables hyperthreading capability on Intel processors Intel HT Technology (HT) is designed to improve performance of multithreaded software products and requires a computer system with a processor supporting HT and an HT-enabled chipset, BIOS and OS. Contact your software provider to determine compatibility. Not all customers or software applications will benefit from the use of HT. See https://www.intel.com/content/www/us/en/gaming/resources/hyper-threading.html for more information.	Checked	Intel CPU with Hyper-Threading Only WMI Name "Hyperthreading" Checked = Enable in WMI
Virtualization Technology (VTx)	Setting	When checked, enables VT on Intel-based systems. NOTE: This setting cannot be disabled when Virtualization Based BIOS Protection is enabled.	Checked	Intel Only Checked = Enable in WMI
Virtualization Technology for Directed I/O (VTd)	Setting	When checked, grants virtual machines direct access to peripheral devices on select Intel-based systems.	Checked	Intel Only Checked = Enable in WMI
SVM CPU Virtualization	Setting	When checked, enables AMD-V and AMD-Vi virtualization features on AMD-based systems. NOTE: This setting cannot be disabled when Virtualization Based BIOS Protection is enabled.	Checked	AMD Only Checked = Enable in WMI
Enhanced Hello Sign-in (G9 or earlier)	Setting	When checked, enables Enhanced Hello Sign- or Enhanced Sign on Security in supported versions of Windows 'ESS' is a required for Microsoft Modern Device security support. Caution: This setting should not be changed with existing Hello enrollments as they will be invalidated.	Unchecked (G9/2022 platforms or earlier)	Select products only Checked = Enable in WMI
Enhanced Sign-In Security (G10 and later)			Checked (G11 / 2024 platforms or later)	
DMA Protection	Setting	When checked, enables DMA redirection using IOMMU for enhanced security. NOTE: Requires Legacy Support disabled and VTd enabled.	Checked	Intel 2019+ AMD 2020+ Checked = Enable in WMI

Feature	Type	Description	Default	Notes
Pre-boot DMA protection	Setting	Secures memory access through DMA to allowed regions prior to OS startup. The following settings are possible: - All PCIe Devices - All PCIe devices with approved exceptions - Thunderbolt™ only (for systems with TBT) - Protects the on-PC Thunderbolt™ Controller - Disable (for systems without TBT)	Notebooks: All PCIe Devices Desktops: Thunderbolt™ only - or - Disable	Intel 2019+ AMD 2021+ This setting requires DMA Protection to be enabled
Measure Additional DMA Settings	Setting	BIOS will measure additional settings values into the TPM. When set to a PCR that BitLocker is using to protect its keys, a BitLocker recovery event will occur if one of the measured settings is changed. The following settings are possible: - Disable - PCR1 - PCR7 NOTE – Help text will appear by clicking '?' within the HP F10 User Interface for this setting: <u>Intel Platforms:</u> (?) When enabled, includes the state of <i>Virtualization Technology (VT-x)</i>, <i>Virtualization Technology for Directed I/O (VT-d)</i>, and the <i>Pre-Boot DMA Protection</i> settings into the measurement for the specified PCR. <u>AMD Platforms:</u> (?) When enabled, includes the state of <i>SVM CPU Virtualization</i>, and the <i>Pre-Boot DMA Protection</i> settings into the measurement for the specified PCR.	G10 or Older Platforms - Disable G11 or Newer Platforms - PCR1	These settings are also available via WMI
Full encryption of main memory (DRAM)	Setting	When checked, the system stores all data to DRAM in an encrypted format.	Checked	Select products only Checked = Enable in WMI
PCI Express x16 Slot 1	Setting	When checked, the PCI Express x16 slot is available for an expansion card. When unchecked, slot is disabled.	Checked	Desktop Only Checked = Enable in WMI
PCI Express x1 Slot 1 (2) (3)	Setting	When checked, the PCI Express x1 slot is available.	Checked	Desktop Only Checked = Enable in WMI

Feature	Type	Description	Default	Notes
PCI Express x4 Slot 1 (2)	Setting	When checked, the PCI Express x4 slot is available.	Checked	Desktop Only Checked = Enable in WMI
PCI Slot 1 (2) (3)	Setting	When checked, the PCI slot is available.	Checked	Select products only Checked = Enable in WMI
M.2 SSD (1) (2)	Setting	When checked, the M.2 slot typically used for NVMe storage modules is available.	Checked	Desktop Only Checked = Enable in WMI
M.2 WLAN/BT	Setting	When checked, the M.2 slot typically used for the WLAN module is available.	Checked	Desktop Only Checked = Enable in WMI
Fast Charge	Setting	When checked, battery charge rate is actively managed by the system using current battery and charger parameters. When unchecked, rate is fixed.	Checked	Notebook Only Checked = Enable in WMI
Power Button Protection	Setting	Disables the power button while off or suspended and the lid is closed to prevent the system turning on when stored (for instance, when in a bag). The following settings are possible: - On Battery Only - Always - Never	On Battery Only	Select products only
Power button delay before sleep or power down	Setting	For products with the power button located on the keyboard, checking this setting will increase the press time required to activate the button to 300 ms. With the setting unchecked the button responds to a keypress in 50 ms.	Checked	Select products only Checked = Enable in WMI

Feature	Type	Description	Default	Notes
Power Button Override	Setting	Sets the time required to hold the power button down for the desktop to turn off, overriding the power button behavior defined by the operating system. The following settings are possible: • Disable • 4 sec • 15 sec	4 sec	Desktop Only
HP Intelligent Hibernate		Hibernation is a very low power state advantageous to use for preserving battery life when the notebook is not being used. A PC in hibernation requires system boot time to get back to work, requiring the user to wait. Intelligent Hibernate solves this problem by using Motion-AI sensing. It intelligently bridges hibernate and wake experiences for increased battery savings and user productivity. Intelligent Hibernate enables multiple hibernate and wake experiences through motion sensing capabilities such as In/Out of bag, Wake Up on Pick Up features. <u>How It Works:</u> In/out of Bag detection: Uses the IMU sensor to detect “in bag” and “out-of-bag” events. Both acceleration and gyroscope data are utilized through a pre-trained AI model and processing on sensor. <u>Wake Up on Pick up:</u> Wakes up automatically from hibernate when “pick up” motion is detected using Motion-AI (IMU only), saving the user 10+ seconds of boot up time. <u>NEW BEHAVIOR FOR G2 Platforms:</u> Intelligent Hibernate will no longer check the Windows Operating System specific power slider position to trigger Hibernation. Instead, the InBag algorithm references different battery drain percentages in S0 and MSC states, to coordinate with Microsoft’s Adaptive Hibernate (AH) logic. This change ensures better alignment with OS-level power management and avoids redundant or conflicting triggers.	Unchecked G2 Platforms: Enabled	Only on G11 models with an NPU On AC Power, the Intelligent Hibernate feature is disabled. Checked = Enable in WMI
Swap Fn and Ctrl (Keys)	Setting	When checked switches functionality between fn and ctrl keys. (This is useful in scenarios where a user works on multiple keyboards where the fn and cntl keys are not in the same position.)	Unchecked	Notebook Only Checked = Enable in WMI
Launch Hotkeys without Fn Keypress	Setting	When checked, allows the fn+fx hot key combinations to be activated by just pressing the fx key, where fx is a key from f1 to f12 (for instance, f4 instead of fn+f4). The 'Auto' setting is available on systems with an LED on the fn key. In Auto mode, fn+shift toggles the fn lock state. • LED off: Checked • LED on: Unchecked	Unchecked or Auto (if available)	Notebook Only Checked = Enable in WMI

Feature	Type	Description	Default	Notes
Deep S4/S5	Setting	The setting adds BIOS capability to reduce power in S4/S5 but requires an AC adapter or a power button press to power up the system. When this feature is enabled and the system is powered off or hibernated, an AC adapter MUST be connected, or a power button should be pressed to power on the system. The following features are unavailable when using Deep S4/S5 Mode: 1. HP Intelligent Hibernation. 2. Power On When Lid is Open. 3. Wake on LAN in Battery Mode. 4. HP Quick Connect. Deep S4/S5 cannot be enabled when the following features are in use: HP Protect and Trace with Wolf Connect	Checked	
Swap Arrow Up/Down and Page Up/Down Function	Setting	When checked, switches functionality between Up / Down keys and Page Up / Page Down keys for platforms with shared keys.	Unchecked	Select products only Checked = Enable in WMI
Special Keys mapped to Fn + keypress	Setting	When checked, maps these key combinations to the legacy keys: • fn+r Break • fn+s Sys Rq • fn+c Scroll lock • fn+w Pause • fn+e Insert NOTE: This setting only applies for systems without these legacy keys.	Unchecked	Select products only Checked = Enable in WMI
Max DC Performance (2019+)	Setting	When checked, allows Intel Turbo Boost Technology to activate when a power adapter is not connected. Previously called <i>Enable Turbo Boost on DC</i> (2018 and older).	Unchecked	Intel Notebook Only Checked = Enable in WMI
Motion sensing cool mode	Setting	This setting is new for 2024 mWKS platforms and 2026 Notebook platforms.	Checked	For supported platforms only Checked = Enable in WMI
Intel Dynamic Tuning	Setting	Manages power and thermal conditions to keep system from overheating. Previously called DPTF.	Checked	Intel Notebook Only Checked = Enable in WMI

Feature	Type	Description	Default	Notes
Sanitization Mode Countdown Timer	Setting	Duration of sanitization mode: 15 – 300 (by 15) in seconds (max 255 for some)	120	Select products only
Pre-Sanitization Mode Countdown Timer	Setting	Delay before sanitization mode starts: 0 – 10 (by 1) in seconds	3	Select products only
USB Type-C Connector System Software Interface (UCSI)	Setting	When checked, allows UCSI to be exposed to the operating system (ACPI table)	Checked	Systems with USB Type-C® ports Checked = Enable in WMI
HP Application Driver	Setting	Provides ACPI structure to enable HP common software application framework. The driver is provided in the latest HP support software which can be downloaded from the web.	Unchecked (2018 and older) Checked (2019+)	Device Manager shows alert if this is enabled without the HP application driver installed. Checked = Enable in WMI
Dynamic Noise Suppression	Setting	(also known as GNA Audio Offload) This feature provides applications with the capability to offload audio (Microphone) processing to a dedicated GNA accelerator, thereby reducing CPU loads and improving battery life.	Checked (for most systems)	Requires driver for function Checked = Enable in WMI
Energy Efficient Turbo	Setting	When checked, allows the system to consider graphics power when deciding on level of CPU turbo frequency.	Unchecked	Select products only Checked = Enable in WMI
AMD DASH	Setting	AMD Remote system management capability.	Unchecked	AMD Only Checked = Enable in WMI

Feature	Type	Description	Default	Notes
Hardware enabled Spectre Variant 2 Mitigation	Setting	This setting enables Single Thread Indirect Branch Predictor (STIBP) functionality in AMD processors.	Unchecked	AMD Only Checked = Enable in WMI
Enable High Resolution mode when connected to a USB-C DP alt mode dock	Setting	Allocate more bandwidth to a USB-C® dock to support the highest resolutions on a DisplayPort monitor attached to it,	Unchecked	Notebook Only Checked = Enable in WMI
Top Cover Function	Setting	Uncheck to disable the top cover functionality for HP Elite Slice.	Checked	HP Elite Slice Only Checked = Enable in WMI
USB Audio Offload	Setting	When checked, allows USB audio processing to be offloaded to dedicated hardware. This improves power efficiency but may increase audio latency and impact compatibility.	Checked	The setting requires driver to function Checked = Enable in WMI

5.7 Built-in Device Options Menu

This menu provides settings for built-in devices on the system.

Table 27 Built-in Device Options Menu features

Feature	Type	Description	Default	Notes
Embedded LAN Controller	Setting	When checked, enables the integrated network controller.	Checked	Select products only Checked = Enable in WMI WMI Name "Embedded LAN controller"

Feature	Type	Description	Default	Notes
Wake On LAN	Setting	Allows the system to wake via Local Area Network (LAN). The following settings are possible: - Disabled - Boot to Network - Boot to Hard Drive - Boot to Normal Boot Order	Boot to Hard Drive	
Dust Filter	Setting	When checked, enables the dust filter reminder. This will prompt you after a period of days specified by the setting below.	Unchecked	Desktop Only Checked = Enable in WMI
Dust Filter Reminder (Days)		Number of days for a reminder to replace the dust filter 15 30 60 90 120 180	60	Desktop Only
LAN Controller Option (1) (2)	Setting	When checked, enables the integrated network controller in the designated rear option slot.	Checked	Select products only Checked = Enable in WMI
Integrated Video	Setting	When unchecked, disables the integrated video device. When not using the integrated video, disabling the integrated video will free some system memory.	Checked	Desktop with discrete graphics card only Checked = Enable in WMI
VGA Boot Device	Setting	The firmware can use only one graphics device when booting up; so when there are multiple graphics devices, this feature selects the graphics controller to use as the primary VGA device during boot-up. - Integrated graphics - Add-in graphics cards (select products only)	Add-in graphics is set as primary	Desktop with discrete graphics card only

Feature	Type	Description	Default	Notes
Video Memory Size	Setting	System memory reserved for video before loading the OS. Settings vary by platform and generation. Examples 2025 and Earlier models: Intel: 64 MB 128 MB 256 MB 512 MB AMD: 256 MB 512 MB - Auto Examples for 2026 and Later models Intel 64 MB, 128 MB, 256 MB, 512 MB AMD - Auto 512 MB 4 GB 8 GB 16 GB 32GB	Intel: 128 MB AMD: Auto	
Preboot Display Clone Mode	Setting	On Intel Desktop systems, Preboot Display Clone Mode will dynamically enable display cloning, to enable viewing in multiple displays. Note: This is done by routing all video to the iGPU during BIOS Post. Once a dGPU driver is loaded, video is routed to the appropriate displays.	Checked	Select desktop products only Checked = Enable in WMI
Graphics	Setting	Set the graphics adapter. The following settings are possible and depend on the model of notebook to determine which are present with the default setting: - Hybrid Graphics - UMA Graphic - Discrete Graphics - Auto (Let OS decide whether hybrid graphics is enabled or disabled).	Hybrid Graphics OR Auto (select products only)	Multiple Graphic Card Notebook Only
Integrated (Front) (Rear) Camera	Setting	When checked, enables the integrated webcams.	Checked	Checked = Enable in WMI

Feature	Type	Description	Default	Notes
HP Camera Privacy Key	Setting	Camera Privacy Key stays in last known state Camera Privacy Key ON after boot or resume	Camera Privacy Key stays in last known state	Camera HW must be in configuration
Internal SD Storage	Setting	When checked, enables integrated SD card controller.	Unchecked	Select products only Checked = Enable in WMI
Fingerprint Device	Setting	When checked, enables fingerprint reader.	Checked	Select products only Checked = Enable in WMI
Touch Device	Setting	When checked, enables the touch screen.	Checked	Select products only Checked = Enable in WMI
Audio Device	Setting	This setting provides a single point of control for the integrated microphone, the internal speakers, and the headphones out. When checked, the operating system visibility of each audio device below is controlled independently. When unchecked, hides all audio devices from the operating systems. The individual audio device settings below are also disabled.	Checked	Checked = Enable in WMI
(Integrated) Microphone	Setting	When unchecked, disables the integrated microphone and microphone jack (if present).	Checked	Notebook Only Checked = Enable in WMI
Microphone	Setting	Set the microphone port state. Possible settings are: • Enable • Disable • Disable and Lock Disable and lock prevents the other audio ports from being remapped to the microphone function in the OS.	Enable	Desktop Only Checked = Enable in WMI
Internal Speakers	Setting	When unchecked, disables the internal speakers. If errors occur during boot-up, the speaker still beeps. See Boot Options / Audio Alerts During Boot for more information.	Checked	Checked = Enable in WMI

Feature	Type	Description	Default	Notes
Headphone Output	Setting	When checked, enables the headphone jack.	Checked	Notebook Only Checked = Enable in WMI
Wake on Voice (WoV)	Setting	When checked, enables the system to wake with voice command.	Checked	Select platforms only Checked = Enable in WMI
Intel Smart Sound	Setting	When checked enables Intel Smart Sound.	Checked	Intel Notebook Only Checked = Enable in WMI
Lock Wireless Button	Setting	Prevent changes to the state of physical wireless enable/disable button.	Unchecked	Notebook Only Checked = Enable in WMI
Wireless Network Device (WLAN)	Setting	When checked, enables integrated 802.11 device.	Checked	Notebook Only Checked = Enable in WMI
Bluetooth	Setting	When checked, enables integrated Bluetooth® device.	Checked	Notebook Only Checked = Enable in WMI
Mobile Network Device (WWAN)	Setting	When checked, enables integrated WWAN device.	Checked	Notebook Only Checked = Enable in WMI
GPS device	Setting	When checked, enables integrated GPS device.	Checked	Notebook Only Checked = Enable in WMI
Mobile Network Device (WWAN) and GPS Combo Device	Setting	When checked, enables integrated WWAN / GPS combo device.	Checked	Notebook Only Checked = Enable in WMI

Feature	Type	Description	Default	Notes
WWAN Quick Connect	Setting	Maintains power to WWAN device to enable faster connections.	Checked	Select products only Checked = Enable in WMI
M.2 USB / Bluetooth	Setting	When checked, enables the USB connection to the M.2 WLAN slot (typically used by Bluetooth if present).	Checked	Desktop Only Checked = Enable in WMI
HP LAN-Wireless Protection	Label			
LAN / WLAN Auto Switching	Setting	When checked, enables automatic switching between embedded WLAN device and embedded LAN controller; disables WLAN when LAN connection is detected.	Unchecked	Checked = Enable in WMI
LAN / WWAN Auto Switching	Setting	When checked, enables automatic switching between embedded WWAN device and embedded LAN controller; disables WWAN when LAN connection is detected.	Unchecked	Notebook Only Checked = Enable in WMI
Wake on WLAN	Setting	When checked, enables the system to wake via WLAN.	Unchecked	Select products only Checked = Enable in WMI
Wake on Bluetooth	Setting	When checked, enables the notebook to wake via BT input devices. Requires Wake on USB to be enabled.	Unchecked	Notebook Only Checked = Enable in WMI
Wake on WiGig	Setting	When checked, enables the notebook to wake via WiGig device.	Unchecked	Notebook Only Checked = Enable in WMI
Collaboration Buttons	Setting	When checked, enables the capacitive controls for volume and connect or disconnect to function.	Checked	Select products only Checked = Enable in WMI
Button Sensitivity	Setting	Controls the touch sensitivity of collaboration buttons. Possible settings are: - Low - Medium - High		Select products only
Hang-up Button Delay	Setting	When checked, hang-up button must be held at least 0.5 sec before activating.	Unchecked	Select products only

Feature	Type	Description	Default	Notes
NFC	Setting	When checked, enable Near Field Communication functionality.	Checked	Select products only Checked = Enable in WMI
Wake on LAN in Battery Mode Wake on LAN on DC mode	Setting	When checked and powered by battery, enables the notebook to wake via LAN.	Unchecked	Notebook Only Checked = Enable in WMI
Fan Always on while on AC Power	Setting	When checked, leaves the fan on while running on AC power.	Unchecked	Notebook Only Checked = Enable in WMI
Customized Fan Control Options	Setting	This setting enables additional system cooling by selecting a value between 1 and 15. A higher number provides more cooling, but also more fan noise. A value of 0 (zero) will revert to the default BIOS fan control settings (no customization). The value selected will remain enabled until changed again from this same menu.	0	Notebook only
Increase Idle Fan Speed(%)	Setting	Controls the minimum fan speed during periods that the fan would normally be off under the control of the desktop thermal sensor. Choose a percentage of the maximum fan speed: 0 –100%.	0	Desktop Only
Boost Converter	Setting	When checked, the notebook draws power from the battery when the system is on AC to give the CPU a momentary performance gain by increasing the overall power available to the CPU.	Checked	Notebook Only Checked = Enable in WMI
Backlit keyboard timeout	Setting	Specifies the timeout period for the keyboard's backlight LEDs. The following settings are possible: • 5 secs • 15 secs • 30 secs • 1 min • 5 min • Never	15 seconds	Select products only
Automatic Keyboard Backlit	Setting	When checked, keyboard backlight level is affected by ambient light level. The keyboard backlight will remain off while in bright environments to save power.	Checked	Select products only Checked = Enable in WMI
RGB Keyboard – System Start-up Lighting Control	Setting	Controls initial state of keyboard lighting for the supported RGB keyboard configurations. Available settings are: Animated, Static, and Disabled	Animated	Select products only

Feature	Type	Description	Default	Notes
Force enable HP Sure View	Setting	When checked, enables HP Sure View's privacy panel by changing the screen brightness	Unchecked	Select products only Checked = Enable in WMI
Magnetic Strip Reader	Setting	When checked, enables the integrated magnetic strip reader	Checked	Select products only Checked = Enable in WMI
Disable Battery on next shut down	Action	When checked, the battery is put in storage mode when the system is next shut down. AC power is required to turn on the system afterwards.	Unchecked	Requires administrator password set Checked = Enable in WMI Old Name "Disable Battery On Next Boot"
RFID Reader	Setting	When checked, enables the RFID reader.	Checked	Select products only Checked = Enable in WMI
RFID Reader Audio Beep	Setting	When checked, audio confirmation is enabled	Checked	Select products only Checked = Enable in WMI
RFID Reader Keyboard Backlight	Setting	When checked, backlight for reader is enabled	Checked	Select products only Checked = Enable in WMI
TILE Deactivate	Setting	When TILE Deactivate is checked, LAN and WLAN location-based services are disabled. Note: TILE features are specific to Intel based Products.	Unchecked	Select NB products only (G8 and G9) Checked = Enable in WMI
TILE Airplane Mode	Setting	RF disabled when selected	Checked	Select products only Checked = Enable in WMI

Feature	Type	Description	Default	Notes
HP Smart Experiences	Setting	Improves camera fidelity with software	Checked	Select products only Checked = Enable in WMI
Wireless Charge in S4/S5	Setting	Keep wireless charger enabled when system is off or hibernating	Unchecked	Select products only Checked = Enable in WMI
Active Pen Wireless Charging	Setting	Enable charging when pen is docked	Checked	Select products only Checked = Enable in WMI
Quick Connect	Setting	When checked, the dock and all connected devices will power on upon user approach, shortening connection latency times. This feature requires additional power. Users who prefer to maximize their PC's battery life may opt to disable this feature.	Unchecked	Notebooks only Checked = Enable in WMI
Quick Connect Notebook Wake	Setting	When checked, the dock will power up the PC from hibernate or off states when it is within proximity of the dock.	Checked	Notebooks only Gray out and no function when Bluetooth is disabled. Hidden when Quick Connect is disabled. Checked = Enable in WMI
Dim Power LED	Setting	Dim the brightness of the power button LED, which may be helpful in low light conditions.	Unchecked	Selected notebooks only Checked = Enable in WMI

5.8 Port Options Menu

The following table describes various setting options for Ports.

Table 28 Port Options Menu features

Feature	Type	Description	Default	Notes
(Left) (Right) (Front) (Rear) (Top) (Bottom) USB Ports	Setting	Enable or disable all USB ports on one side of the system (legacy and Type-C).	Checked	Checked = Enable in WMI
(Left) (Right) (Front) (Rear) USB Port (1) (2) (3) (4) (5) (6) (7) (8)	Setting	Enable or disable a specific USB port. NOTE: When looking at the ports (and in horizontal orientation for desktops), count ports from bottom to top, then left to right.	Checked	Desktop Only Checked = Enable in WMI
Docking USB Ports	Setting	When unchecked, disables USB ports connected through the docking connector.	Checked	Notebook Only
USB Legacy Port Charging	Setting	When checked, enables the USB Type-A charging port to charge devices during hibernation or shutdown.	Checked	Checked = Enable in WMI
Disable Charging Port in sleep/off if battery below (%)	Setting	Prevent charging port from providing power to external devices if the system itself is below a certain battery threshold. Possible settings are: 10, 20, 30, 40, 50, 60, 70, 80, 90, 100.	10	Notebook Only
(Front) (Rear) USB Type-C Downstream Charging	Setting	When unchecked, the system will not power Type-C devices in the off states.	Checked	Desktop Only Checked = Enable in WMI
Thunderbolt Type-C Ports	Setting	When checked, enables integrated Thunderbolt™ ports. NOTE: Older systems included additional Thunderbolt™ settings in this menu. Starting in 2019 these options have moved to a separate Thunderbolt™ Options menu.	Checked	Select products only Checked = Enable in WMI
USB Type-C Ports	Setting	Disables power delivery at USB-C port when notebook PC is turned off.	Checked	Select products only Checked = Enable in WMI
Accessory USB Port	Setting	When checked, enables the accessory USB port.	Checked	Desktop Workstations Only Checked = Enable in WMI

Feature	Type	Description	Default	Notes
(Rear) Option Port (1) (2)	Setting	When checked, enables the identified option port without regard to which option type is installed.	Checked	Select desktop products only Checked = Enable in WMI
Option Port (1) (2) - HDMI 1.4 Mode	Setting	When checked, enables additional bandwidth for DisplayPort® over Type-C to support higher graphics resolutions.	Unchecked	Select products only Checked = Enable in WMI
Media Card Reader	Setting	When checked, enables the integrated media card reader.	Checked	Notebook & AiO Only Checked = Enable in WMI
Media Card Reader/SD_RDR USB	Setting	When checked, enables the media card reader connector (labeled SD_RDR typically) on a desktop.	Checked	Desktop Only Checked = Enable in WMI
SATA (0) (1) (2) (3) (4) (5)	Setting	When checked, allows the system to access a device attached to the SATA port.	Checked	Desktop Only Checked = Enable in WMI
Serial Port (A, B, C, D, C/D, E/F)	Setting	When checked, enables the specified serial ports.	Checked	Desktop Only Checked = Enable in WMI
Serial Port Voltage (A) (B) (C) (D) (E) (F)	Setting	Powered Serial port voltage selection on RPOS units that include this feature. Possible settings are: 0 Volts 5 Volts 12 Volts	0 Volts	Retail Point of Sale Systems Only Checked = Enable in WMI
Smart Card	Setting	When checked, enables integrated Smart Card slot.	Checked	Notebook Only Checked = Enable in WMI
Smart Card Power Savings	Setting	When checked, enables the power-saving feature of the Smart Card reader, thus not maintaining a session when the card is removed.	Checked	Notebook Only Checked = Enable in WMI
Cash Drawer Port	Setting	On select Retail Point of Sale systems, this controls whether the cash drawer port can be activated or not.	Enable	Retail Point of Sale Systems Only

Feature	Type	Description	Default	Notes
				Checked = Enable in WMI
Restrict USB Devices	Setting	When some devices are restricted, the system disables the ports at boot-up where a restricted device is installed. That port is disabled until the next boot. Port configuration is <i>not</i> changed on insertion. The following settings are possible: - Allow all USB Devices - Allow only keyboard and mouse - Allow all but storage devices and hubs	Allow all USB Devices	
HUB LAN Controller	Setting	RPOS Only	Checked	Engage One Pro G2 only
HUB Wake On LAN	Setting	RPOS Only	Unchecked	Grayed out if HUB LAN Controller is disabled
Dual Type-A 3.2 Gen1 Flex/IO ports	Setting	Enables or disables the port(s). Default is checked (Ports On)	Checked	Desktop only

5.9 Network Settings

The following table describes various options for Network Settings.

Table 29 Network Settings Options Menu features

Feature	Type	Description	Default	Notes
Extended DHCP Timeout	Setting	When enabled, allows DHCP waits to establish connection quicker.	Checked	Select products only Checked = Enable in WMI
Network TCP Buffer Size	Setting	This setting specifies the network TCP buffer size. Recommended Network TCP Buffer Size: 32768 *65536 (Default) 2097152	65536	HP G8 (2021) and later platforms.

5.10 Power Management Options Menu

The following table describes various setting options for Power Management Options.

Table 30 Power Management Options Menu features

Feature	Type	Description	Default	Notes
Runtime Power Management	Setting	When checked, enables the processor to run at lower frequencies (P-states) when higher performance is not needed. When unchecked, the processor always runs at maximum frequency.	Checked	Select products only Checked = Enable in WMI
Extended Idle Power States	Setting	When checked, enables the processor to rest in lower power states (C-states) when idle.	Checked	Select products only Checked = Enable in WMI
S5 Maximum Power Savings	Setting	When checked, minimizes system power consumption while in the S5 (off) state. NOTE: Windows 10 with Fast Startup enabled powers off to the S4 (suspend to disk) state.	Unchecked	Desktop Only Checked = Enable in WMI
SATA Power Management	Setting	When checked, enables the SATA bus to enter low power states when idle.	Checked	Desktop Only Checked = Enable in WMI
PCI Express Power Management	Setting	When checked, enables PCI Express bus to enter low power states when idle.	Checked	Desktop Only Checked = Enable in WMI
PCIe Speed Power Policy (PSPP)	Setting	When checked, allows the system to lower PCIe link speeds when not on AC to save battery power.	Checked	AMD Notebook Only Checked = Enable in WMI
AMD Core Performance Boost	Setting	AMD Core Performance Boost (CPB), also known as AMD Turbo Core, is a technology that allows AMD processors to dynamically increase their clock speeds beyond the base frequency when thermal headroom and power limits allow. This leads to improved performance during demanding tasks, while maintaining lower power consumption and heat generation during less intensive workloads.	Checked	AMD only Checked = Enable in WMI
Power On from Keyboard Ports	Setting	When checked, allows the desktop to turn on by pressing a key on the keyboard, when the keyboard is plugged in to a port marked with the keyboard symbol.	Unchecked	Desktop Only Checked = Enable in WMI
Unique Sleep State Blink Rates	Setting	When checked, while the desktop is in the S4 power state, the power LED periodically blinks four times with a pause. Unchecked, the desktop does not blink at all in S4 (the same as S5, power off) This also affects S3 blink behavior. When checked, the desktop power LED periodically blinks three times with a pause, unchecked it blinks once per period.	Unchecked	Desktop Only Checked = Enable in WMI

Feature	Type	Description	Default	Notes
Wake when Lid is Opened	Setting	When checked, opening the lid wakes the notebook from sleep mode	Unchecked	Selected 2021 and 2022 Notebooks Only Checked = Enable in WMI
Wake when AC is Detected	Setting	When checked, allows the system to resume from sleep when AC power is detected		Selected 2021 and 2022 Notebooks Only Checked = Enable in WMI
Wake on USB	Setting	When checked, allows the system to resume from sleep when a USB input device is triggered (such as mouse movement or keyboard keypress).	Checked	Selected 2021 and 2022 Notebooks Only Checked = Enable in WMI
Power Control	Setting	When checked, enables the notebook to support power management applications such as IPM+ that help enterprises reduce power costs by intelligently managing the battery usage of the notebook.	Unchecked	Notebook Only Checked = Enable in WMI
Battery Health Manager	Setting	Sets charging policy based on optimizing for battery life or for battery duration. The possible settings are: - Maximize Battery Health Management - Let HP Manage My Battery Health (Default) - Minimize Battery Health Management	Let HP manage my battery charging	Notebook Only
Modern Standby	Setting	Low power standby mode. This mode replaces the traditional ACPI S3 sleep and S4 hibernation states.	Enable	Selected 2021 products only

Feature	Type	Description	Default	Notes
Charger Ship Mode Deep S4/S5	Setting	Reduces battery power consumption in S4/S5 states. A different BIOS capability – ‘Battery Ship Mode’ can reduce power in S4/S5 but requires an AC adapter to power up the system. Deep S4/S5 leverages hardware enhancements and firmware that provides improved battery life in S4/S5 and does not require an AC Adapter to power up the system. Power button press can power up the system. The system enters the Deep S4/S5 state after remaining on battery and in S4/S5 power state continuously for 24 hours. Deep S4/S5 does impact other BIOS settings and features. When system is in Deep S4/S5 state, the below BIOS settings and features are not active: 1. HP Intelligent Hibernate. 2. Power On When Lid is Open. 3. Wake on LAN in Battery Mode. 4. HP Quick Connect. 5. Single Sign on using Fingerprint sensor Deep S4/S5 cannot be enabled when the following settings are enabled: • HP Protect and Trace with Wolf Connect.	Checked	When HP Wolf Connect is installed and active on the PC, Charger Ship Mode cannot be enabled. Checked = Enable in WMI

5.11 Remote Management Options Menu (Intel Only)

The following table describes various setting options for Remote Management Options.

Table 31 Remote Management Options Menu features

Feature	Type	Description	Default	Notes
Intel Active Management Technology (AMT)	Setting	This setting controls the Intel Active Management Technology (AMT) remote manageability features. When unchecked, the network based remote management functionality is disabled.	Checked (unchecked for mobile workstations)	Intel Only Checked = Enable in WMI Old Name “Active Management (AMT)”
USB Key Provisioning Support	Setting	When checked, enables AMT provisioning using a USB storage device. NOTE: Unavailable when AMT is disabled.	Unchecked	Intel Only Checked = Enable in WMI
USB Redirection Support	Setting	When checked, enables support for storage redirection through USB NOTE: Intel AMT must be correctly provisioned NOTE: Unavailable when AMT is disabled.	Checked	Intel Only Checked = Enable in WMI

Feature	Type	Description	Default	Notes
Unconfigure AMT on next boot	One time action	When applied, reset AMT configuration options on next boot. The following actions are possible: - Do Not Apply - Apply NOTE: Unavailable when AMT is disabled.	Do Not Apply	Intel Only
SOL Terminal Emulation Mode	Setting	Specifies the Serial Over Lan (SOL) terminal emulation mode. The following settings are possible: - ANSI - VT100 NOTE: Unavailable when AMT is disabled.	ANSI	Intel Only
Show Unconfigure ME Confirmation Prompt	Setting	When checked, requires user confirmation when unconfiguring Intel Management Engine. NOTE: Unavailable when AMT is disabled.	Checked	Intel Only Checked = Enable in WMI
Verbose Boot Messages	Setting	When checked, report additional information when a boot message is displayed. NOTE: Unavailable when AMT is disabled.	Unchecked	Intel Only Checked = Enable in WMI
Watchdog Timer	Setting	When checked, enables Watchdog Timers. NOTE: Unavailable when AMT is disabled.	Unchecked	Intel Only Checked = Enable in WMI
OS Watchdog Timer (min.)	Setting	Sets OS Watchdog Timer (minutes). Possible values are from 5 to 25. NOTE: Unavailable when AMT is disabled.	5	Intel Only Checked = Enable in WMI
BIOS Watchdog Timer (min.)	Setting	Sets BIOS Watchdog Timer (minutes). Possible values are from 5 to 25. NOTE: Unavailable when AMT is disabled.	5	Intel Only Checked = Enable in WMI
CIRA Timeout (min.)	Setting	Client Initiated Remote Access timeout. Possible values are from 1 to 4 minutes or never. NOTE: Unavailable when AMT is disabled.	1 min	Intel Only Checked = Enable in WMI

5.12 MAC Address Pass Through (Notebook Only)

The following table describes various settings for the Host Based MAC Address menu.

Table 32 MAC Address Pass Through Menu features

Feature	Type	Description	Default	Notes
Host Based MAC Address	Setting	Can be set to: - Disable - System - Custom (Setting to Custom allows MAC address to be modified.)	System Address	Notebook Only
MAC ADDRESS	Setting	Configure a custom MAC address. Shows the current factory and system MAC addresses as well.	Factory MAC Address	Notebook Only

5.13 Thunderbolt™ Options

The following table describes various settings for configuring Thunderbolt™ ports. There still remains a setting in Port Options to turn the Thunderbolt™ port on or off.

Table 33 Thunderbolt Options Menu features

Feature	Type	Description	Default	Notes
Thunderbolt Mode	Setting	When checked, enables Thunderbolt™ connections on the Type-C port. When unchecked: - Disables Thunderbolt™ connections on the Type-C port - Disables PCIe Tunneling on USB4 connections	Checked	Only on Intel platforms with Thunderbolt support or on ZBook Power G10 A Checked = Enable in WMI

5.14 Remote HP PC Hardware Diagnostics Settings

Table 34 Remote HP PC Hardware Diagnostics Features

Feature	Type	Description	Default	Notes
Diagnostic Download URL	Setting	HP / Custom URL.	HP	WMI Name "Remote HP PC Hardware Diagnostics Use Custom Download Url" HP = Disable in WMI Custom URL = Enable in WMI
Custom Download Address	Setting	Location of Remote Diagnostics, if not obtained from the HP server.		WMI Name "Remote HP PC Hardware Diagnostics Custom Client Download Url"
Custom Upload Address	Setting	Custom location to upload Diagnostic logs.		WMI Name "Remote HP PC Hardware Diagnostics Custom Client Upload Url"
Username	Setting	(Optional) Username to access custom Diagnostic location.		WMI Name "Remote HP PC Hardware Diagnostics Upload Server Username"

Feature	Type	Description	Default	Notes
Password	Setting	(Optional) Password to access custom Diagnostic location.		WMI Name "Remote HP PC Hardware Diagnostics Upload Server Password"
Scheduled Execution	Setting	Allow Remote HP PC Diagnostics to run on a set schedule: • Enable • Disable	Disable	WMI Name "Remote HP PC Hardware Diagnostics Scheduled Execution Enabled"
Frequency	Setting	Select the frequency for scheduled execution of Remote HP PC Hardware Diagnostics: • Daily • Weekly • Monthly	Weekly	WMI Name "Remote HP PC Hardware Diagnostics Scheduled Execution Frequency"
Execute On Next Boot	Setting	Enable or disable the execution on next boot. The Flag will be disabled after the diagnostics have run: • Enable • Disable	Disable	WMI Name "Remote HP PC Hardware Diagnostics Execute On Next Boot"
Last Execution Result	Action	Displays the result of the last Remote HP PC Diagnostics execution		WMI Name "Remote HP PC Hardware Diagnostics Last Execution Status"

6 UEFI Drivers

Main

Security

Advanced

UEFI Drivers



HP Computer Setup

This feature restarts the system into the 3rd Party Option ROM Management application. You can get to this application directly by pressing F3 during startup.

⇒ **3rd Party Option ROM Management**

7 Settings Not in F10 Menu

These settings are BIOS controlled via WMI but do not have an option or setting in the F10 menu.

Table 35 Features not in F10 Menu

Feature	Description	Default	Notes
Privacy Panel	For notebooks equipped with Privacy Panel, press fn+f2 to enable or disable the privacy panel feature. Use fn+f5 and fn+f6 to decrease or increase the privacy panel brightness.	Disabled	For select privacy panel notebooks only.
Lock System and Power Off	For systems equipped with Firmware Lock and with a Local Access Key present, open the (ESC) Startup Menu and select the 'Lock System and Power Off' option. When prompted, provide the Sure Admin PIN to shut down and lock the system. NOTE: Beginning in 2024, this setting is only available on PCs with Enterprise Security Edition.	Disabled	
Lock System on Next Boot	For systems equipped with Firmware Lock and with a Local Access Key present, software within the operating system can use this setting to enable Firmware Lock on the next boot. • Disable • Enable NOTE: Beginning in 2024, this setting is only available on PCs with Enterprise Security Edition.	Disable	

Feature	Description	Default	Notes
Lock System Authorization	For systems equipped with Firmware Lock and with a Local Access Key present, this allows an administrator to control what level of BIOS authorization is needed to change the “Lock System on Next Boot” setting above. - Always Required - Never Required - Always Required Except First Usage NOTE: Beginning in 2024, this setting is only available on PCs with Enterprise Security Edition.	Always Required	
Permanent Disable Absolute Persistence Module Set Once	This allows customers to permanently disable the Absolute Persistence Module on their platform. - No - Yes	No	
Permanently Disable HP Sure Run (Set Once)	This allows customers to permanently disable HP Sure Run on their platform. Note that this impacts the persistence claims for HP Protect & Trace.	No	
Permanently Disable Remote Device Management (Set Once)	This allows customers to permanently disable the Wolf Protect & Trace features on their platform. - No - Yes (This setting was introduced in the second half of 2021)	No	
Permanently Disable HP Cloud Management (Set Once)	This allows customers to permanently disable the use of HP cloud based manageability tools (e.g., Wolf Controller & TechPulse) to provision / rotate / remove the cryptographic keys used with features such as Wolf Protect & Trace, Sure Admin and Sure Recover, in a simple straightforward manner. - No - Yes (This setting was introduced in the second half of 2021)	No	
Secure Erase Completion Date	The last date a disk was erased by Secure Erase. (Display Only)	Blank	

Feature	Description	Default	Notes
Secure Erase Completion Status	The completion status of Secure Erase on the last disk erased. (Display Only)	Blank	
Secure Erase Hard Disk Model Number	The model of the last disk erased by Secure Erase. (Display Only)	Blank	
Secure Erase Hard Disk Serial Number	The serial number of the last disk erased by Secure Erase. (Display Only)	Blank	
Secure Erase Drive Info	This displays the drives that are present in the system, their technology (e.g., NVMe, SATA, eMMC, etc.) and the command that would be issued in Secure Erase to perform the most secure erase the BIOS is able to perform.	Varies	For example: SAMSUNG MZVLC2T0HBLD-00BH7;S82GNE0Y800368;NVMe;Crypto Erase
Secure Platform Management Counter	This displays the current anti-replay counter for secure platform management commands. (Display Only)	0	
Secure Platform Management Key Endorsement Certificate	This is a write only WMI used to set the certificate from which the BIOS should extract the key endorsement key. Reads will return an empty string.	Blank	
Secure Platform Management Key Endorsement Key	This displays a base64 encoded string value of the key endorsement public key extracted from the certificate that was set. (Display Only)	Blank	
Secure Platform Management Signing Key	This displays a base64 encoded string value of the signing public key. (Display Only)	Blank	
Secure Platform Management Usage Bitmask	This displays a bitmask of features currently using Secure Platform Management. (Display Only) Bit <0> = HP Sure Run Bit <1> = HP Sure Recover Bit <2> = HP Sure Admin Bit <15-3> = reserved (0)	0x0000	
Secure Platform Management Version	This displays the version of Secure Platform Management support. (Display Only)	1.00	
Enhanced BIOS Authentication Mode Version	This displays the version of Sure Admin's Enhanced BIOS Authentication Mode support. (Display Only)	1.01	

Feature	Description	Default	Notes
Enhanced BIOS Authentication Mode Local Access Key 1	This displays the current base64 encoded string value of Sure Admin's Local Access Key (and can be used to set it). In cases where a Key Management Server is being used, there is also information about its URL appended.	Blank	
Enhanced BIOS Authentication Mode Settings Anti-Replay Counter	This displays the anti-replay counter value used by Sure Admin's when changing values via WMI. In the case of 'settings' it is expected that some remediation solutions would want to resend the last change so the anti-replay comparison is greater than or equal. (Display Only)	0	
Enhanced BIOS Authentication Mode Actions Anti-Replay Counter	This displays the anti-replay counter value used by Sure Admin's when changing values via WMI. In the case of 'actions' it is expected that the action should never be accepted more than once so the anti-replay comparison is strictly greater than. (Display Only)	0	
HP EpSC GUID	This displays the unique GUID assigned to HP's Endpoint Security Controller. (Display Only)	Unique value	
HP EpSC Attestation Certificate	This displays the base64 encoded string value of the Attestation certificate in HP's Endpoint Security Controller. (Display Only)	Unique per system	
HP EpSC Device Authentication Certificate	This displays the base64 encoded string value of the Authentication certificate in HP's Endpoint Security Controller. (Display Only)	Unique per system	
HP EpSC Device ID Certificate	This displays the base64 encoded string value of the Device ID certificate in HP's Endpoint Security Controller. (Display Only)	Unique per system	
OS Recovery Agent Public Key	This displays the base64 encoded string value of the public key currently used to verify the Sure Recover Agent. (Display Only)	HP Factory Image Key	
OS Recovery Image Public Key	This displays the base64 encoded string value of the public key currently used to verify the OS Image to be used by Sure Recover. (Display Only)	HP Factory Image Key	

Feature	Description	Default	Notes
Custom Logo	This displays whether or not a custom logo has been configured on the system. (Display Only) - Not Present - Present	Not Present	
DriveLock Enabled Drives	This displays a list of the drives on which DriveLock has been enabled. (Display Only)	Blank	Value formatted as: M.2 SSD: MTFDKBA1T0QGN-1BN1AABHA-UTXHU0153JMGEU
Ready to disable MS UEFI CA Key	This displays whether or not it is possible to disable the UEFI CA based on whether any key components are signed with the UEFI CA and won't work if we disable it. (Display Only) - Not Ready - Ready	Depends on system	
Custom Keys Image Verification State	This displays whether there are custom secure boot keys installed and whether or not they are valid. (Display Only) - No Custom Keys - Fail - Success	No Custom Keys	
Intel(R) Thunderbolt Retimer FW version	This displays the version of the firmware being used for the Intel Thunderbolt Retimer.	Varies	
Intel(R) Thunderbolt Retimer FW Model ID	This displays the version of the firmware being used for the Intel Thunderbolt Retimer.	Varies	
Battery Safety Mode	This setting is enabled automatically if a known bad HP battery is detected the firmware. If enabled the system will disconnect the battery and require AC power to be used. A message will be displayed when booting stating that the battery should be replaced. - Disable - Enable	Disable	
Cirrus Discrete Amp Calibration Data	This displays calibration data used by the audio system. (Display Only)	Varies	
BIOS Build Version	This displays the BIOS Build Version. (Display Only)	Varies	
Force Cold Boot			
Force Default IP Configuration			

Feature	Description	Default	Notes
Force to Run Discrete Only Mode			
HBMA Priority List			
Manufacturing Programming Mode			
SureStart Production Mode			
MS Digital Marker			
Pre-boot HBMA Support			
Remote HP PC Hardware Diagnostics Last Execution Time Stamp			
Single NIC Operation (Disable All Other NICs when HBMA is active on one NIC)			
Windows HBMA Support			
HBMA Custom MAC Address			
HBMA Factory MAC Address			
HBMA System MAC Address			

8 Computer Notifications

8.1 Introduction

Platforms that support HP PC Commercial BIOS have various mechanisms to indicate errors that occur during Power-On-Self-Test (POST). The notifications can take several forms, such as:

- Blinks and Beeps
- On screen notifications that include the following:
 - Preboot messages (BIOS)
 - Pop-up messages within the OS

8.2 Blink and Beep Codes

Some system errors prevent the use of the video screen; instead, the system provides error information through blink codes using LED lights. The LED light used depends on the system type (notebook or desktop). The codes are presented in a sequence. For desktop, this means red blinks followed by white blinks. Audible long and short beeps accompany red or white blinks, respectively. Additional details may be found in the system’s Maintenance and Service Guide.

The following table describes the meaning of critical blink codes.

Table 36 Computer notifications

Notebook		Desktop		Description
CAP / NUM	Battery LED	Red with long beeps	White with short beeps	
2		2	2	The main area of BIOS has become corrupted, and there is no recovery binary image available.
8		2	3	The HP Endpoint Security Controller policy requires the user to enter a key sequence (Sure Start 2.0).
	White and Amber blinking	2	4	The HP Endpoint Security Controller is recovering the BIOS firmware. Because it takes some time to load the firmware image and enable video, this blink code is necessary. (Sure Start).
3		3	2	The HP Endpoint Security Controller has timed out waiting for BIOS to return from memory initialization (memory failure).
4		3	3	The HP Endpoint Security Controller has timed out waiting for BIOS to return from graphics initialization.
5		3	4	The system board displays a power failure (crowbar).
		3	5	The CPU is not detected or is unsupported.
		3	6	The CPU does not support an enabled feature (typically this applies only to TXT).
7	1	5	2	The HP Endpoint Security Controller cannot find valid firmware.

8.3 Toast (Pop-Up) Messages

Onscreen notification can involve pop-up (toaster) messages. These describe several events involving USB Type-C® ports. Note that these messages within the OS require native support in the operating system or that HP notifications software be installed.

Table 37 Pop-up messages

Event	Code	Message	Detail
Power Adapter Accepted: Matches capabilities to charge while in S3, S4 or S5 power states.	1	Title: USB Type-C® Connector Text: "For full performance, connect a higher capacity power adapter."	A user plugs in a power adapter that is too small to operate the system while the device is turned on. The adapter could be used to charge in sleep mode or when the computer is turned off.
Power adapter rejected: Upstream power flow is not supported	2	Title: USB Type-C® Connector Text: "Charging system via adapter plugged into the USB port is not supported."	A user plugs in an adapter that requests power in which is not supported. (Cypress controller)
Connected device requests more power than can be supplied	3	Title: USB Type-C® Connector Text: "USB device requesting more power than system can provide." <i>Display system charging capability</i>	A user plugs in a device that requires more power than can be provided by the system.
Balance downstream power for charging from multiple USB ports	4, 5	Title: USB Type-C® Connector Text: "Charging from multiple USB ports may have limited support."	A user has plugged in an adapter to both a USB Type-A port and a USB Type-C® port (or into two USB Type-C® ports), and the system is not capable of charging both at full capacity while system is running.
The attached dock cable is inadequate to handle the needed power load	6	Title: USB Connector Text: "For full performance, connect higher capacity USB cable to dock." <i>Display capabilities of the cable</i>	A user plugs a cable connecting the dock to the system that is inadequate to power the system and charge the battery simultaneously.
Power adapter rejected: Provider and consumer mismatch	7	Title: USB Connector Text: "The power adapter is not compatible with this system."	The user has inserted an adapter that is not compatible with the HP system (from a 3 rd party vendor that is not supported.)

9 Appendix A: UEFI

9.1 What is UEFI?

Unified Extensible Firmware Interface (UEFI) defines the interface between the operating system and platform firmware during the boot, or start-up process. Compared to BIOS, UEFI supports advanced preboot user interfaces.

The UEFI network stack enables implementation on a richer network-based OS deployment environment while still supporting traditional PXE deployments. UEFI supports both IPv4 and IPv6 networks. In addition, features such as Secure Boot, enable platform vendors to implement an OS-agnostic approach to securing systems in the preboot environment.

The HP ROM-Based Setup Utility (RBSU) functionality is available from the UEFI interface with additional configuration options.

9.2 Introduction

The HP UEFI System Utilities are embedded in the system ROM. The UEFI System Utilities enable a wide range of configuration activities, including:

- Configuring system devices and installed options
- Enabling and disabling system features
- Displaying system information
- Selecting the primary boot controller or partition
- Configuring memory options
- Launching other pre-boot environments, such as the Embedded UEFI Shell and Intelligent Provisioning

9.3 Benefits of UEFI

- Abstracts platform from OS and decouples development
- Utilizes a modular driver model and CPU-independent option ROMs
- Remains modular and extensible and provides OS-neutral value-add
- Includes an OS loader that can keep the same as underlying hardware change
- Supports larger drives over 2 TB with GPT partition

9.4 Overview of UEFI Boot Process

The purpose of the UEFI interfaces is to define a common boot environment abstraction for use by loaded UEFI images, which include UEFI drivers, UEFI applications, and UEFI OS loaders. UEFI allows the extension of platform firmware by loading UEFI driver and UEFI application images. When UEFI drivers and UEFI applications are loaded they have access to all UEFI-defined runtime and boot services.

There are two sets of services in UEFI:

- **Boot Services** - UEFI applications (including OS loaders) must use boot services functions to access devices and allocate memory. These services are not available when the OS is running.
- **Runtime Services** - The primary purpose of runtime services is to abstract minor parts of the hardware implementation of the platform from the OS.

These services are present when OS is running.

9.5 The UEFI Forum

For more information, contact the Unified Extensible Firmware Interface (UEFI) Forum. It is a world-class nonprofit industry standards body that works in partnership to enable the evolution of platform technologies.

The UEFI Forum champions firmware innovation through industry collaboration and the advocacy of a standardized interface that simplifies and secures platform initialization and firmware bootstrap operations. Both developed and supported by representatives from more than 200 industry-leading technology companies, UEFI specifications promote business and technological efficiency, improve performance and security, facilitate interoperability between devices, platforms and systems, and comply with next-generation technologies.

10 Appendix B: Firmware Update

10.1 Updating System Firmware with the HP Firmware Update and Recovery Application

(Windows Operating Systems only)

Current firmware updates for HP commercial platforms include the HP Firmware Update and Recovery tool (HpFirmwareUpdRec.exe). This utility starts the firmware update process when run with the correct firmware source files for the target platform. Firmware types supported by this utility include the BIOS, the ME firmware (*Intel only*), and USB Type-C® PD (power delivery) controller firmware. When the utility is run in Windows, it identifies the compatible firmware files in local storage and then invokes a series of flash updates after triggering a system reboot.

Before 2018, the firmware update tool was HP BIOS Update and Recovery (HpBiosUpdRec.exe), which uses the specific BIOS binary included in the Softpaq as an input (for example, P70_010102.bin). This tool operates in a similar fashion to the newer HP Firmware Update and Recovery tool.

For 2018 and later systems, the firmware source files required for updating within BIOS Setup (F10) menus must be extracted from the .bin and .inf files included in the release Softpaq. The Firmware Update and Recovery application must be used to extract the various firmware binary files to use the Update System and Supported Device Firmware Using Local Media action in BIOS Setup. For earlier platforms, only the appropriate BIOS binary file from the Softpaq is required.

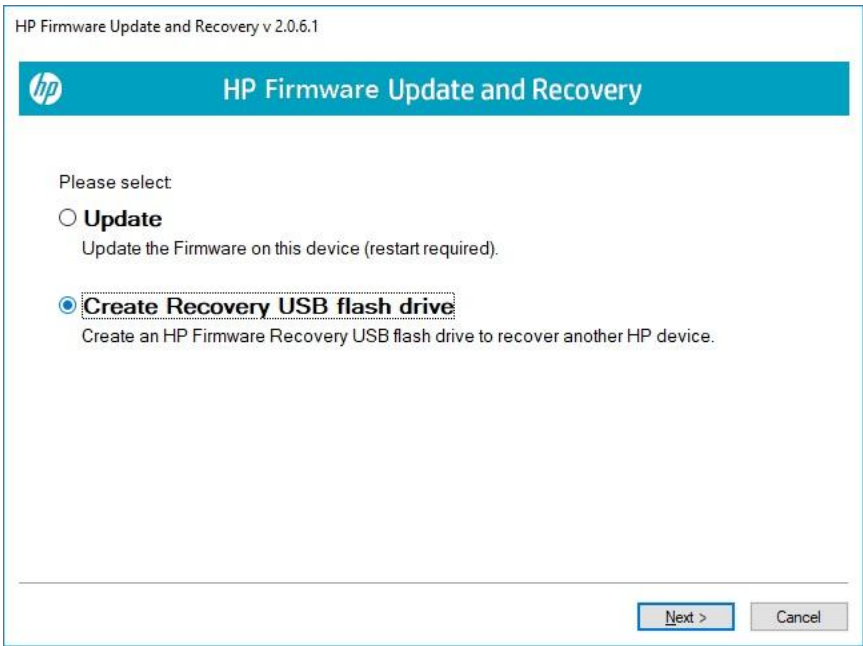
10.2 Firmware Update and Recovery Documentation

The HP Firmware Update and Recovery tool is component of the HP SoftPaq firmware update bundle. Included with the SoftPaq is a user guide that provides command line options for a variety of options. Along with the user guide, release notes are also provided within the package. These SoftPaq updates are available on the HP website (Support) for each platform.

10.3 USB Recovery Key Creation

If the system BIOS has been corrupted and the device will not boot, another device can be used to create an HP Firmware Recovery USB Key that can be used to recover it. The device used to create the recovery key does not have to be compatible with the BIOS image.

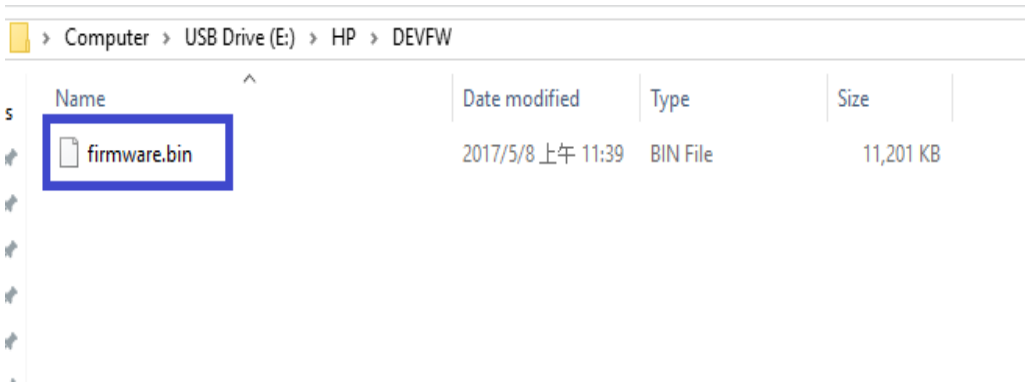
- Run the **HpFirmwareUpdRec** or **HpBiosUpdRec** application. The main options menu is shown with the following options: 'Update' and 'Create Recovery USB flash drive'.



- Select **Create Recovery USB flash drive** and then select **Next**.
- The application will prompt to insert a USB flash drive if the system does not see a USB flash drive.
- 
Select a USB flash drive and click **Next**.
- 
Upon completion, the application will display a message that the recovery flash drive was created successfully. Click **Finish** to close the wizard.

The files can also be manually copied to the EFI partition of the hard drive to support emergency recovery. For 2018 and later the HpFirmwareUpdRec utility extracts the correct binaries from the .bin and .inf files and saves the individual components on the USB key in the HP\DEVFW folder.

NOTE: To recover a device with the flash drive, connect AC power and follow the previous on-screen instructions.



10.4 HpFirmwareUpdRec Log File

By default, a log file is created in the same folder with the executable file.

- If the `-l` command line option is used, the log file will be written to the supplied file path. If it is a relative path, it will be placed under that path.
- If the log file cannot be created in the executable folder, it will be created in the first available system temporary folder location, usually “C:\Users\<username>\AppData\Local\Temp” in Windows.

10.4.1 Custom Logo Support

NOTE: Operates in Silent Mode only, will not update firmware.

Installation:

- Command Line: HpFirmwareUpdRec.exe -e<logo filename>
- Custom Logo file will be written to BIOS. Check the log file for success or error.
 - **NOTE:** File must be JPEG format, maximum size 32k (32,768) bytes.
- If BIOS password is set, you must provide the password file.

Command-line option only, silent mode, not shown in usage display. Other options ignored. System will not be restarted.

Removal:

- Command Line: HpFirmwareUpdRec.exe -x
- Logo image will be removed from BIOS.
- If BIOS password is set, you must provide the password file.

Command line option only, silent mode, not shown in usage. Other options ignored. System will not be restarted.

- Custom logo support

Return Code	Name	Description
0	SUCCESS	No error
1	LOAD_ERROR	Error reading image file
2	INVALID_PARAMETER	File name missing
3	UNSUPPORTED	Not supported by BIOS
4	INVALID_FILE	File not found or invalid
26	SECURITY_VIOLATION	BIOS password not provided or incorrect
99	UNKNOWN	Unknown error occurred, see log file

10.4.2 Command-line Usage

Table 38 Custom logo support: command-line usage

Option	Comments
-f	Folder Path. Specifies a folder path for tool to scan both inf and bin files. Default Path: Application folder where HPFirmwareUpdRec.exe exists.
-s	Silent mode. Runs without any user interaction or output.
-ni	Non-Interactive mode. Not compatible with -s, -t, -o, -c, -x, -v, or -silentPostpone.
-p	Specifies an encrypted password file created with the HPQPswd Utility. For more information on HPQPswd Utility, please reference Section 8.
-r	Do not reboot after flash. The result code is SUCCESS_REBOOT (0xBC2) when this option is used.
-h	Creates HP_TOOLS partition if not present. On a GPT formatted system with native UEFI boot, this option is ignored. On MBR, the partition is not created if it already exists. If unable to create partition, exits with error code.
-b	If BitLocker with TPM is in use, automatically suspend it.
-w	Outputs firmware.bin
-delay	Force EFI flash even if Seamless Firmware Update mode is supported
-silentPostpone	Triggers Silent Mode even if '-s' is not provided and restart/postpone window will be launched at the end of the update
-t	Postpone time in minutes. Default is 10 minutes. Max value is 60 minutes.
-o	Number of postpone occurrences allowed. Default is 3. Max value is 5.
-disableToast	Disable showing a toast notification in the Windows Notification Centre to indicate completion of firmware update. Restart/postpone window will be launched at the end of an update. For Non-Interactive Mode, this option is always enabled.
-?	Displays all public command line options. Shows this same usage dialog that appears if an invalid command line is detected. This option overrides all other command options.
-c	Copies update files to EFI partition BIOS\Current folder, without setting update ready flag. Implies -s as well, so files are copied with only a validation of the ROM family.
-logfile_path	Specifies log file path. This overwrites any existing log file at <i>logfile_path</i> ; i.e., new log records are not appended to an existing file, but instead new log file is created at that path. Default is HPFirmwareUpdRec.log in the executable folder.
-elogofile	Runs in silent mode. Install and enable an enterprise logo image. No BIOS update or restart. The result code is 0 on success.
-x	Runs in silent mode. Remove and disable any existing enterprise logo image. No BIOS update or restart. The result code is 0 on success.
-v	Ignores version. Eliminates version comparison warnings in UI mode.
-bloverride	Skip check for BitLocker – do not prompt or suspend

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